Practical No : 01

AIM : Demonstrate the use of various features of Excel that are required to do the data analysis.

Result Analysis

Open Excel

Select a blank sheet

Suppose, The data set we are having **initially**

		Marks			
Roll no	**Name**	**Sub 1**	**Sub 2**	**Sub 3**	**Sub 4**
1	Vikas Yadav	25	35	60	56
2	Amit Mishra	45	60	55	78
3	Sunita Yadav	56	42	40	84
4	Priyanka Gupta	45	35	20	38
5	Anita Shetty	37	40	35	90
6	Ranjan Yadav	56	56	35	60
7	Ankita Mishra	34	30	60	61
8	Vaibhav Yadav	89	78	90	89
9	Soniya Gupta	24	78	35	45
10	Sunil Shetty	45	56	40	89

Operation 1

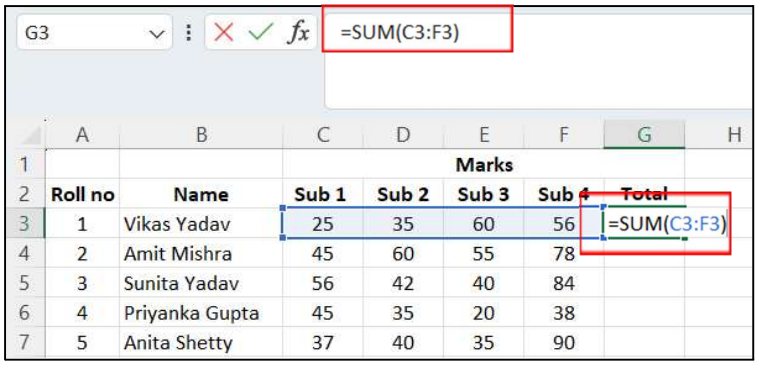
We wanted to have the **total marks** of the students

Use the **Sum formulas**

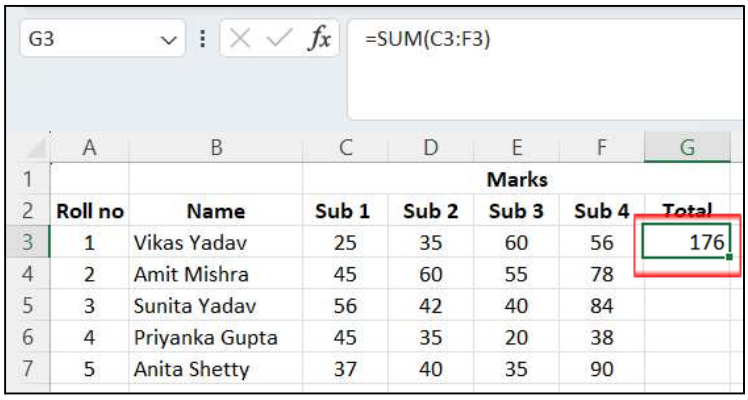
1) **=SUM(start cell: end cell)**

   Or

2) **=(cell1 + cell2 + cell3 + …..)**

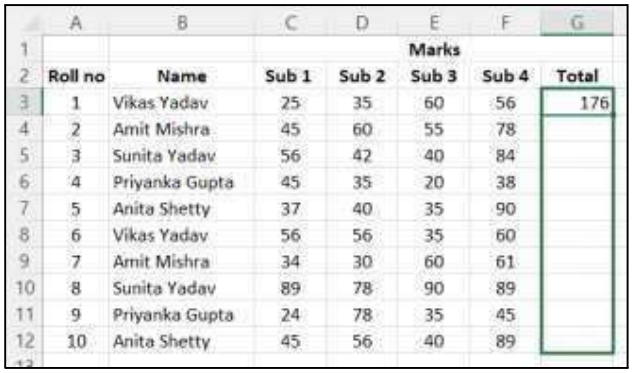


You can see the **sum of marks from sub 1 to sub 4** is present in the **total** mark column



Now for all the others, I don't need to write the formula again and again.

Just select the cell where you have applied the formula and drag the cursor to the column to which you want to apply the same formula.



After the release of the cursor, you can see the total marks of all student in their respective total column

	A	B	C	D	E	F	G
1					Marks		
2	Roll no	Name	Sub 1	Sub 2	Sub 3	Sub 4	Total
3	1	Vikas Yadav	25	35	60	56	176
4	2	Amit Mishra	45	60	55	78	238
5	3	Sunita Yadav	56	42	40	84	222
6	4	Priyanka Gupta	45	35	20	38	138
7	5	Anita Shetty	37	40	35	90	202
8	6	Vikas Yadav	56	56	35	60	207
9	7	Amit Mishra	34	30	60	61	185
10	8	Sunita Yadav	89	78	90	89	346
11	9	Priyanka Gupta	24	78	35	45	182
12	10	Anita Shetty	45	56	40	89	230

Operation 2

Now, let's derive Percentage of the Students As

percentage is calculated as

(mark obtained in all subjects/total marks of all subjects) * 100

=(G3/400)*100

C	D	E	F	G	H
		Marks			
b 1	Sub 2	Sub 3	Sub 4	Total	Percentage
5	35	60	56	176	=(G3/400)*100
5	60	55	78	238	

	A	B	C	D	E	F	G	H
1					Marks			
2	Roll no	Name	Sub 1	Sub 2	Sub 3	Sub 4	Total	Percentage
3	1	Vikas Yadav	25	35	60	56	176	44
4	2	Amit Mishra	45	60	55	78	238	59.5
5	3	Sunita Yadav	56	42	40	84	222	55.5
6	4	Priyanka Gupta	45	35	20	38	138	34.5
7	5	Anita Shetty	37	40	35	90	202	50.5
8	6	Ranjan Yadav	56	56	35	60	207	51.75
9	7	Ankita Mishra	34	30	60	61	185	46.25
10	8	Vaibhav Yadav	89	78	90	89	346	86.5
11	9	Soniya Gupta	24	78	35	45	182	45.5
12	10	Sunil Shetty	45	56	40	89	230	57.5

Operation 3

As the percentage is showing in decimal points what if I want the percentage to be a whole number only

Apply formula

=ROUNDUP(Cell no, digits to round up)

=ROUNDUP(H3, 0)

C	D	E	F	G	H	I
		Marks				
b 1	Sub 2	Sub 3	Sub 4	Total	Percentage	% in Whole No.
25	35	60	56	176	44	=ROUNDUP(H3, 0)
45	60	55	78	238	59.5	

You can see the percentage in whole number only

H	I
Percentage	% in Whole No.
44	44
59.5	60
55.5	56
34.5	35
50.5	51
51.75	52
46.25	47
86.5	87
45.5	46
57.5	58

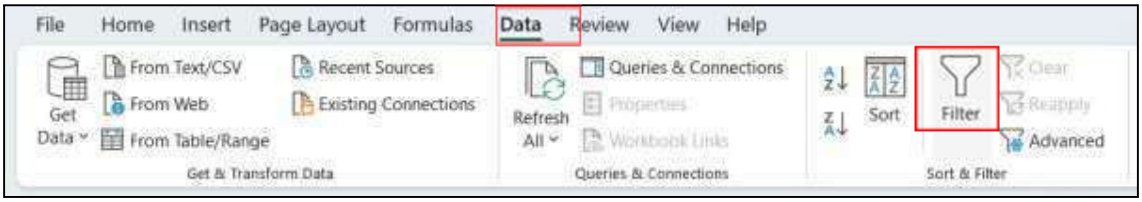
Operation 4

Suppose, you want to **mark** the students who are failing **in the individual subjects**, The criteria for a student to be filed in a student if he is **marks less than 40**

Step 1: Select your table

	A	B	C	D	E	F	G	H	I
1					Marks				
2	Roll no	Name	Sub 1	Sub 2	Sub 3	Sub 4	Total	Percentage	% in Whole No.
3	1	Vikas Yadav	25	35	60	56	176	44	44
4	2	Amit Mishra	45	60	55	78	238	59.5	60
5	3	Sunita Yadav	56	42	40	84	222	55.5	56
6	4	Priyanka Gupta	45	35	20	38	138	34.5	35
7	5	Anita Shetty	37	40	35	90	202	50.5	51
8	6	Ranjan Yadav	56	56	35	60	207	51.75	52
9	7	Ankita Mishra	34	30	60	61	185	46.25	47
10	8	Vaibhav Yadav	89	78	90	89	346	86.5	87
11	9	Soniya Gupta	24	78	35	45	182	45.5	46
12	10	Sunil Shetty	45	56	40	89	230	57.5	58
13									

Step 2: go to the '**Data**' tab and in the Data tab select the '**filter**' option

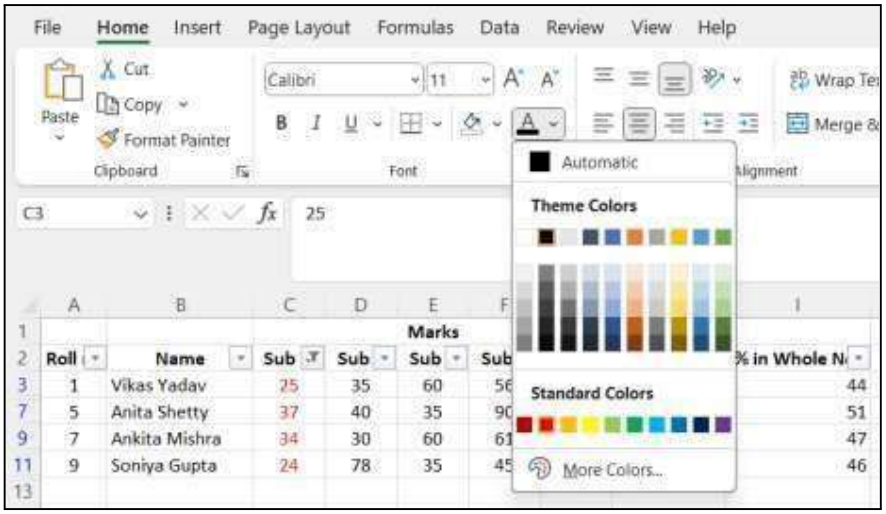


After clicking on the filter option you can see the table heading a down arrow symbol in the column that is filtering options

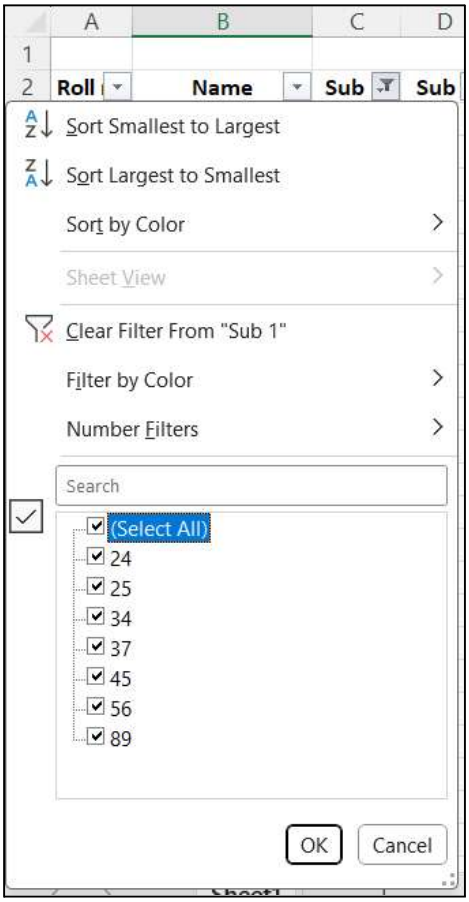
	A	B	C	D	E	F	G	H	I
1					Marks				
2	Roll	Name	Sub	Sub	Sub	Sub	Tota	Percentag	% in Whole N
3	1	Vikas Yadav	25	35	60	56	176	44	44
4	2	Amit Mishra	45	60	55	78	238	59.5	60

Step 3: click the arrow of the 'sub 1' column uncheck the select all option select only the marks that match the fail criteria and click on OK

You will be only able to see the students who have less than 40 marks in sub 1
Select the marks of sub1 and mark them in red color this student fails in sub 1 because they got less than 40 marks



Now, again click on the sub 1 column arrow select all, and click on ok, and all the records will be visible. This is important if you forget then you will operate only on the selected data not



on the entire data.

After doing this do the same form sub 2, sub 3, and sub 4 columns of marks

	A	B	C	D	E	F	G	H	I
1					Marks				
2	Roll	Name	Sub	Sub	Sub	Sub	Tota	Percentag	% in Whole N
3	1	Vikas Yadav	25	35	60	56	176	44	44
4	2	Amit Mishra	45	60	55	78	238	59.5	60
5	3	Sunita Yadav	56	42	40	84	222	55.5	56
6	4	Priyanka Gupta	45	35	20	38	138	34.5	35
7	5	Anita Shetty	37	40	35	90	202	50.5	51
8	6	Ranjan Yadav	56	56	35	60	207	51.75	52
9	7	Ankita Mishra	34	30	60	61	185	46.25	47
10	8	Vaibhav Yadav	89	78	90	89	346	86.5	87
11	9	Soniya Gupta	24	78	35	45	182	45.5	46
12	10	Sunil Shetty	45	56	40	89	230	57.5	58

After filtering data for failed students,
We can say the the students who are marked in red color in the sub mark column failed in that particular subject

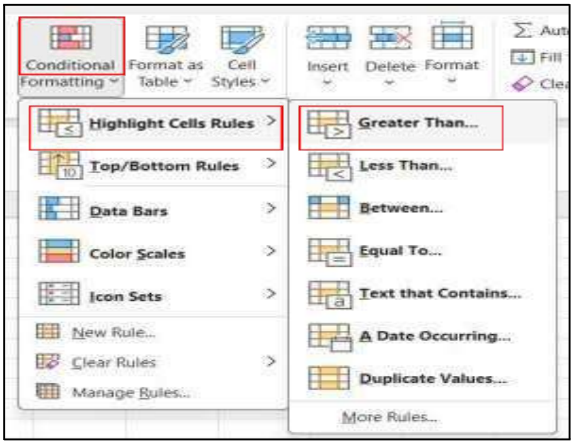
Suppose, you want to remove that data filtering option again select the data table go to the Data tab menu, and unselect the filter option and you will see the filtering option is now removed from columns headings

Operation 5
Let Highlight the percentage marks which are greater than 80%

Select the cells

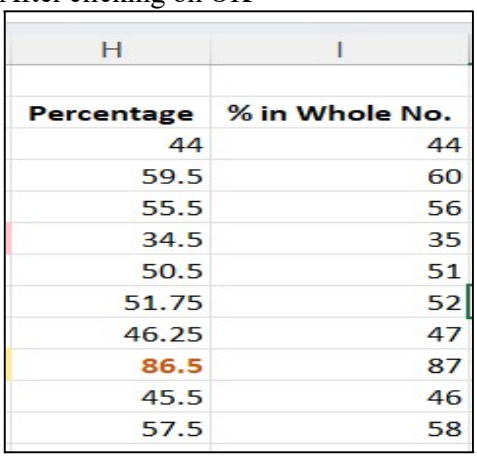
H	I
Percentage	% in Whole No.
44	44
59.5	60
55.5	56
34.5	35
50.5	51
51.75	52
46.25	47
86.5	87
45.5	46
57.5	58

Go to Home tab → Conditional Formatting → Highlight cell rules → Greater than



I have formatted as font color - orange and style bold

After clicking on OK



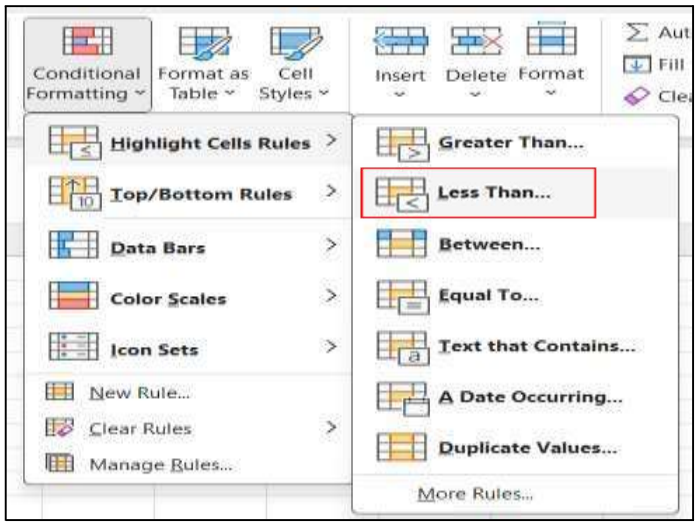
H	I
Percentage	% in Whole No.
44	44
59.5	60
55.5	56
34.5	35
50.5	51
51.75	52
46.25	47
86.5	87
45.5	46
57.5	58

Operation 6

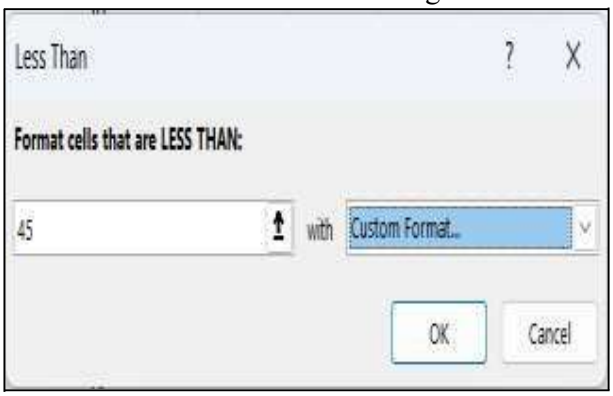
Let Highlight the percentage marks which are lesser than 45%

Again select the cells

Go to Home tab → Conditional Formatting → Highlight cell rules → Greater than



Less than 45% with custom formatting font-color red and style bold



H	I
Percentage	% in Whole No.
44	44
59.5	60
55.5	56
34.5	35
50.5	51
51.75	52
46.25	47
86.5	87
45.5	46
57.5	58

Operation 7

Now,

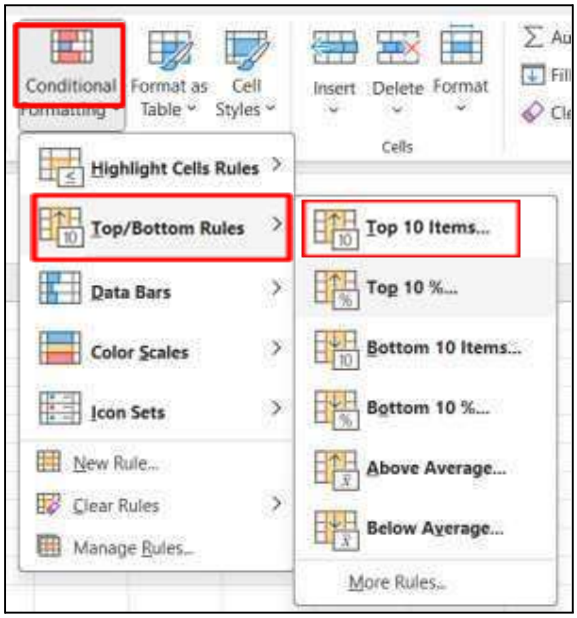
Let look for the Highest and Lowest scorer from the total mark obtained by the students

We will mark the cell as eYellow for the highest scorer and red as the lowest scorer

Select the cells

Marks			
Sub 3	Sub 4	Total	Percentage
60	56	176	44
55	78	238	59.5
40	84	222	55.5
20	38	138	34.5
35	90	202	50.5
35	60	207	51.75
60	61	185	46.25
90	89	346	86.5
35	45	182	45.5
40	89	230	57.5

Go to Home tab → Conditional Formatting –. Top/Bottom Rules → Top 10 Items



The below following dialog bax will open give the no. of time you want and with which formatting style

I gave top **1** with **Yellow fill and Dark Yellow text** click on OK

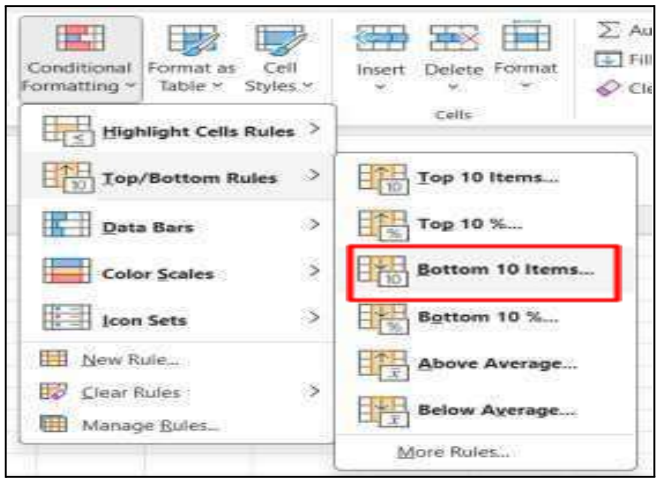
And you come to see the highest scorer in yellow

E	F	G	H	I
Marks				
Sub 3	Sub 4	Total	Percentage	% in Whole No.
60	56	176	44	44
55	78	238	59.5	60
40	84	222	55.5	56
20	38	138	34.5	35
35	90	202	50.5	51
35	60	207	51.75	52
60	61	185	46.25	47
90	89	346	86.5	87
35	45	182	45.5	46
40	89	230	57.5	58

And do for Lowest scorer

Select the cells

Go to Home tab → Conditional Formatting –. Top/Bottom Rules → Bottom 10 Items

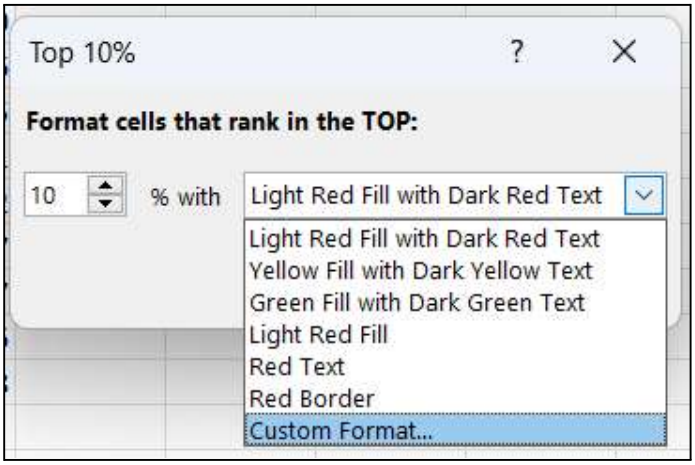


	A	B	C	D	E	F	G
1					Marks		
2	Roll no	Name	Sub 1	Sub 2	Sub 3	Sub 4	Total
3	1	Vikas Yadav	25	35	60	56	176
4	2	Amit Mishra	45	60	55	78	238
5	3	Sunita Yadav	56	42	40	84	222
6	4	Priyanka Gupta	45	35	20	38	138
7	5	Anita Shetty	37	40	35	90	202
8	6	Ranjan Yadav	56	56	35	60	207
9	7	Ankita Mishra	34	30	60	61	185
10	8	Vaibhav Yadav	89	78	90	89	346
11	9	Soniya Gupta	24	78	35	45	182
12	10	Sunil Shetty	45	56	40	89	230

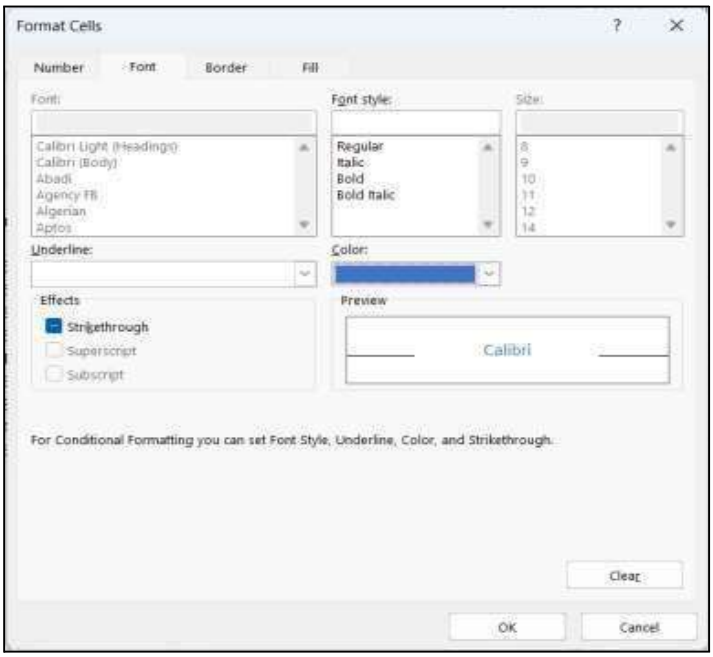
Operation 8

Let find the top 10% scorer from each subject from above lessons

But now we will use custom formatting



I have choose color blue and fill light blue



Press on Ok and Again OK

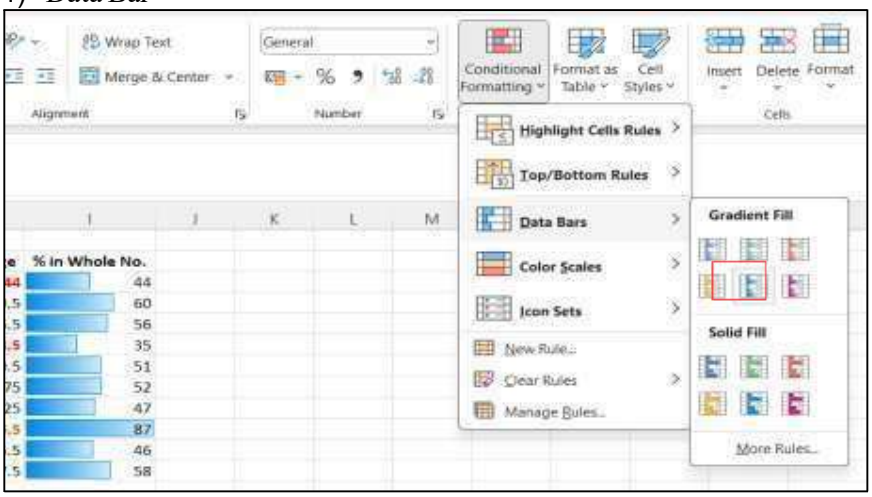
And you will see
Top 10% of each sub in blue cell formatting

	A	B	C	D	E	F	G
1					Marks		
2	Roll no	Name	Sub 1	Sub 2	Sub 3	Sub 4	Total
3	1	Vikas Yadav	25	35	60	56	176
4	2	Amit Mishra	45	60	55	78	238
5	3	Sunita Yadav	56	42	40	84	222
6	4	Priyanka Gupta	45	35	20	38	138
7	5	Anita Shetty	37	40	35	90	202
8	6	Ranjan Yadav	56	56	35	60	207
9	7	Ankita Mishra	34	30	60	61	185
10	8	Vaibhav Yadav	89	78	90	89	346
11	9	Soniya Gupta	24	78	35	45	182
12	10	Sunil Shetty	45	56	40	89	230

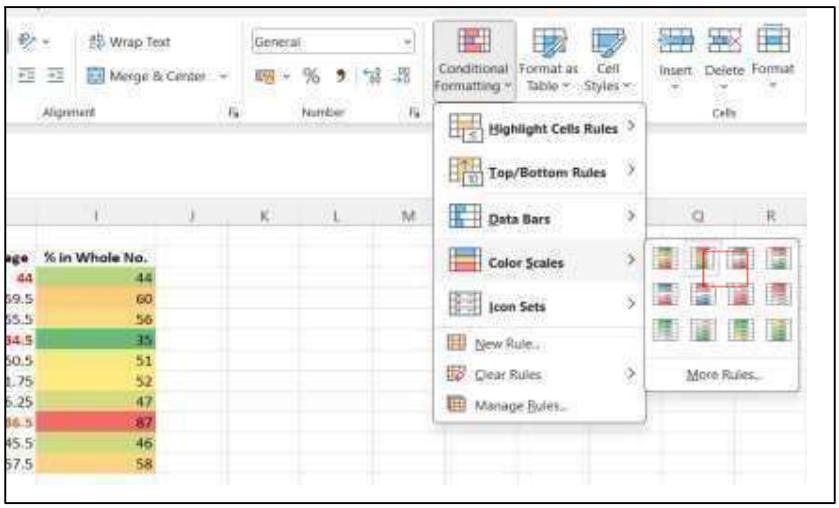
Operation 9

Let go through the this three option this options used for formatting the cells

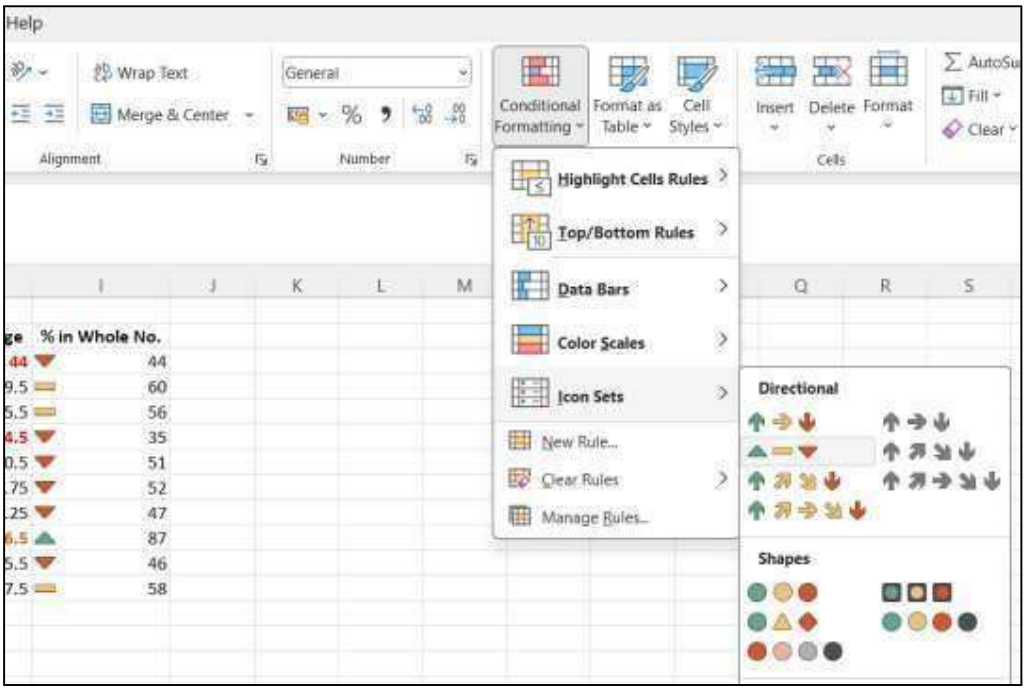
1) Data Bar



2) Color Scales



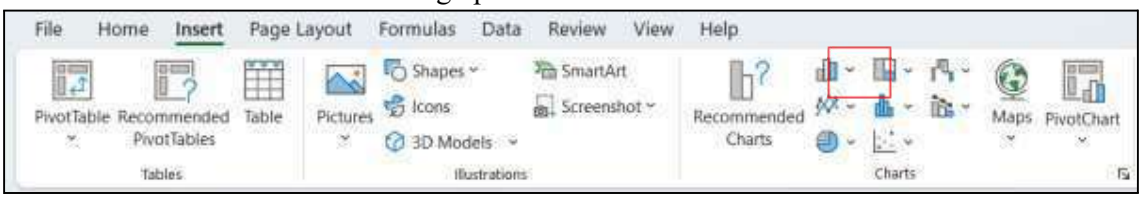
3) Icon sets



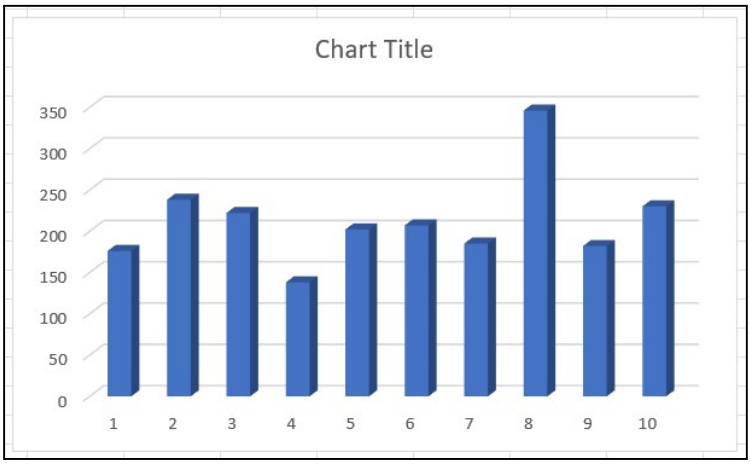
From clear rules Options you can clear the conditional formatting rules from the cells

Operation 10
Data
Visualization

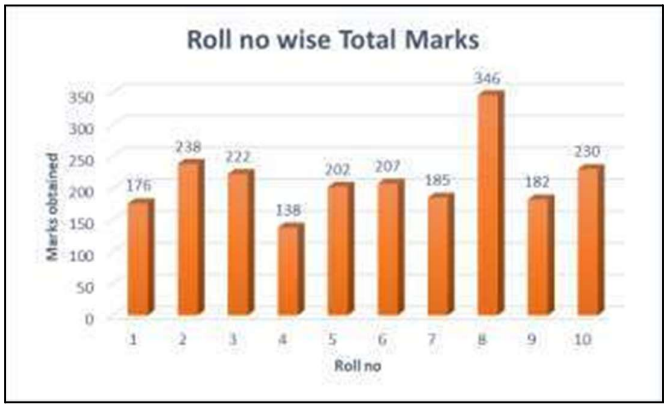
Now,
Select total Marks column
Go to Insert → Charts ans select bar graph



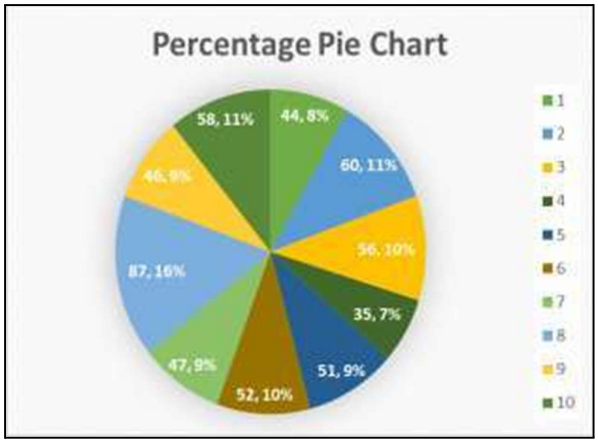
This charts will appear



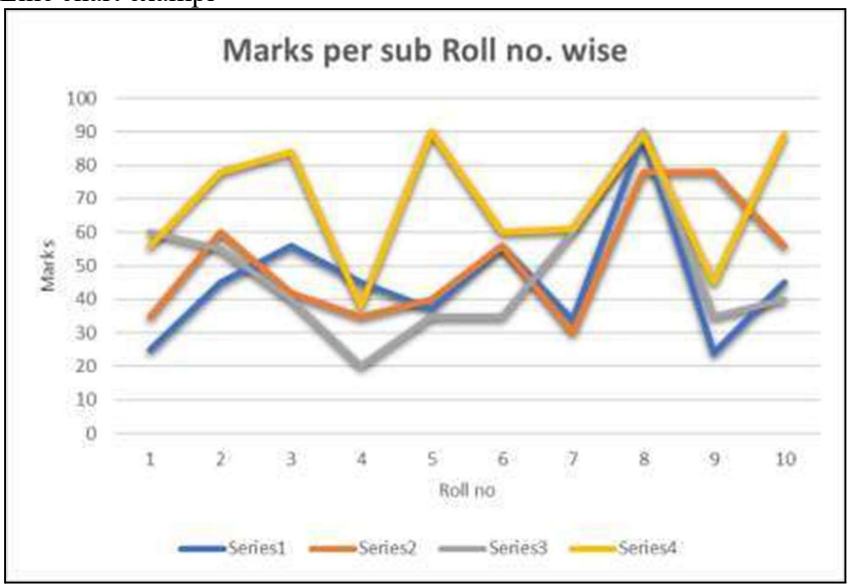
Format the graph as per our preference



Another chart example



Line chart exampl

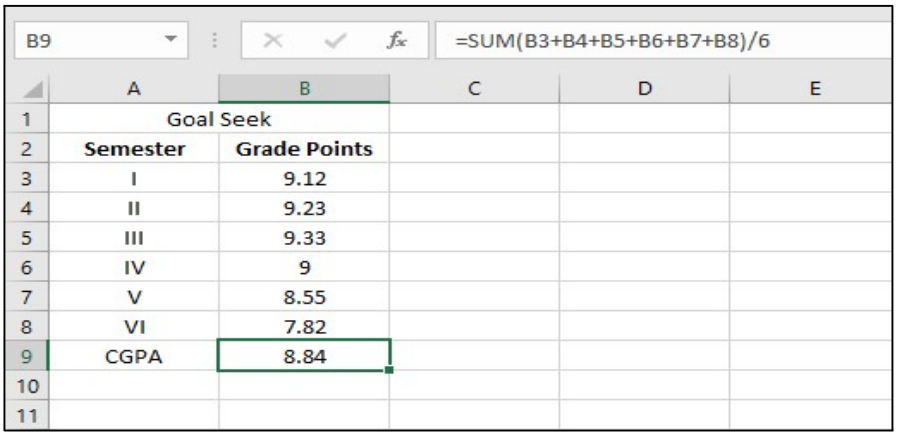


Practical No : 02

AIM : Demonstrate what-if analysis using goal seek, scenario manager and data table for your own data.

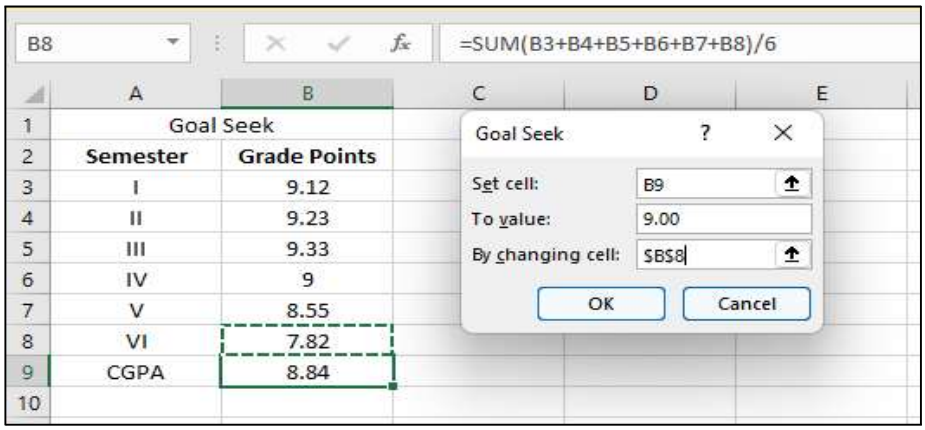
Goal Seek : This will find the right input value.

Example 1 - Semester wise SGPA scored



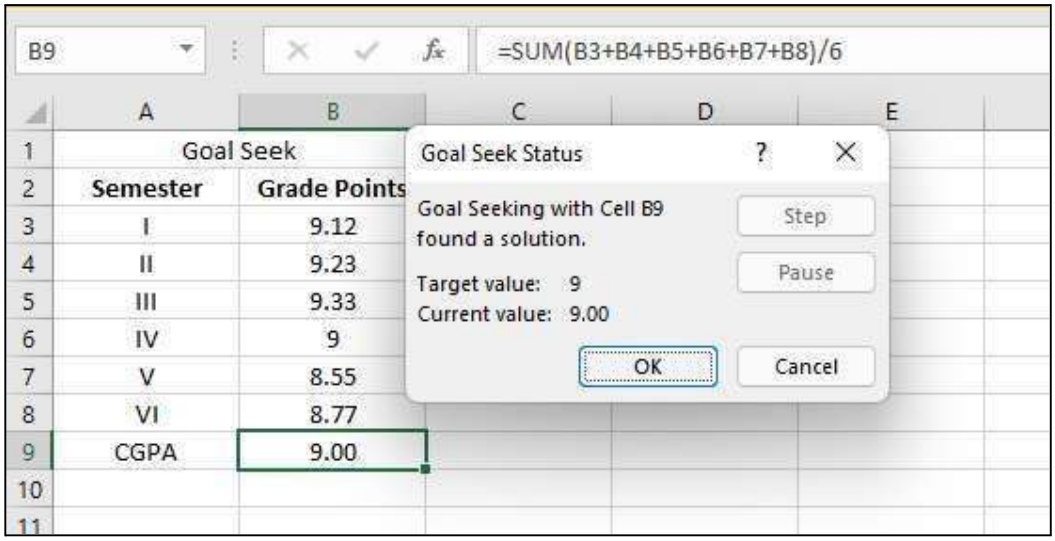
Problem Statement - Suppose you are in semester 6 and after completion of your degree to wanted to achieve the aggregate CGPA of 9.5 the you want to know how much you have to score in SEM 6 to get the the aggregate CGPA of 9.5

Go to Data → What-If Analysis → Goal Seek..



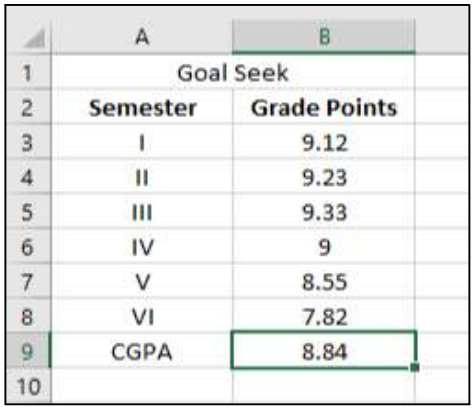
Set cell → final CGPA cell
To value → CGPA score you want to achieve
By changing cell → Sem 6 Grade point cell

Click on OK



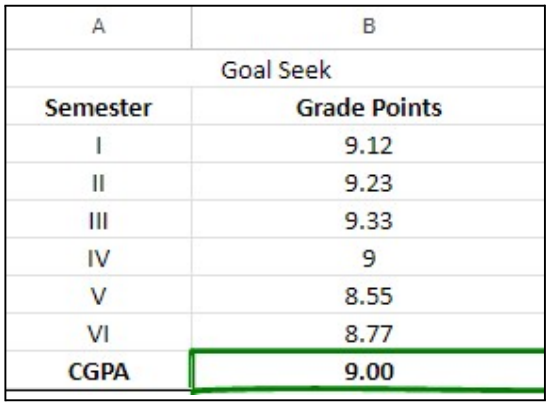
The result will be

Before



	A	B
1	Goal Seek	
2	Semester	Grade Points
3	I	9.12
4	II	9.23
5	III	9.33
6	IV	9
7	V	8.55
8	VI	7.82
9	CGPA	8.84
10		

After



A	B
Goal Seek	
Semester	Grade Points
I	9.12
II	9.23
III	9.33
IV	9
V	8.55
VI	8.77
CGPA	9.00

Conclusion

If you want to achieve an Aggregate of 9.00 CGPA then i need to score 8.77 SGPA in sem 6

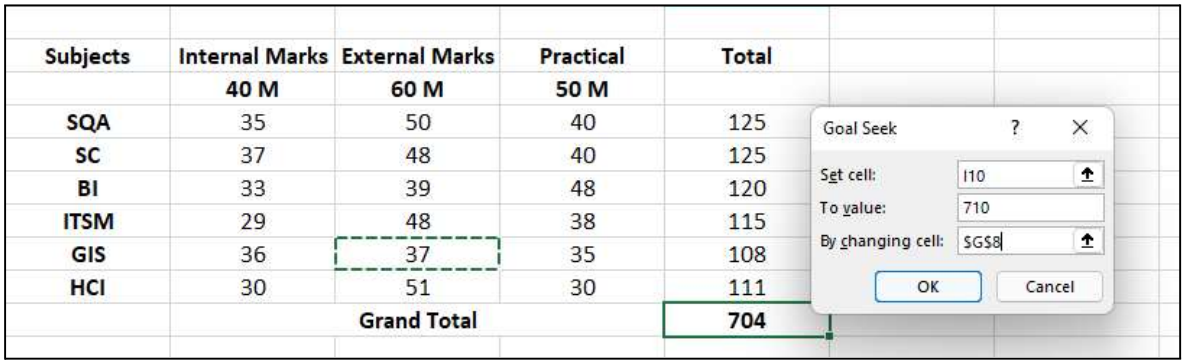
Example 2 - Subject and Total

Enter the below data

Subjects	Internal Marks	External Marks	Practical	Total
	40 M	60 M	50 M	
SQA	35	50	40	125
SC	37	48	40	125
BI	33	39	48	120
ITSM	29	48	38	115
GIS	36	37	35	108
HCI	30	51	30	111
		Grand Total		704

Apply Goal Seek

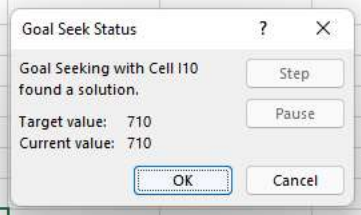
Go to → Data → What-If analysis → Goal Seek..



Subjects	Internal Marks	External Marks	Practical	Total
	40 M	60 M	50 M	
SQA	35	50	40	125
SC	37	48	40	125
BI	33	39	48	120
ITSM	29	48	38	115
GIS	36	37	35	108
HCI	30	51	30	111
		Grand Total		704

Set cell → total marks cell
To value → marks you want to achieve
By changing cell → Grand Total cell

Subjects	Internal Marks	External Marks	Practical	Total
	40 M	60 M	50 M	
SQA	35	50	40	125
SC	37	48	40	125
BI	33	39	48	120
ITSM	29	48	38	115
GIS	36	43	35	114
HCI	30	51	30	111
		Grand Total		710



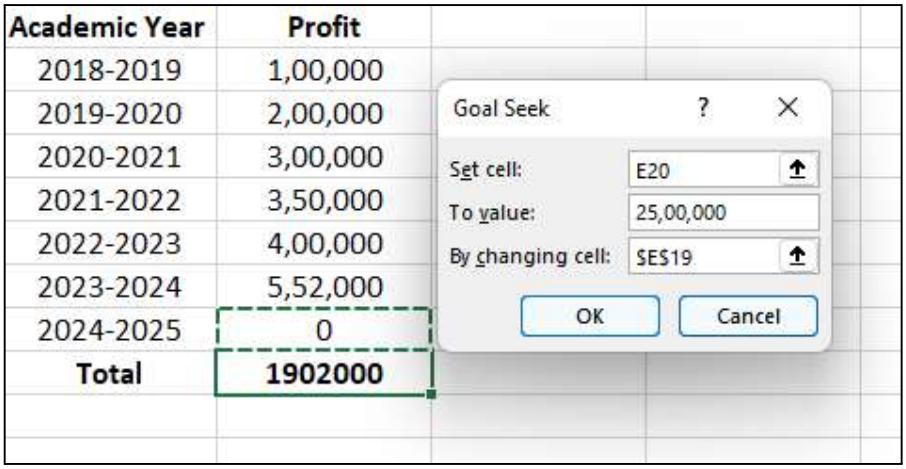
The result will be

Example 3 - Academic year

You have given the profit of a Company ABC for respective Academic Year
You want to know how much you have to make profit in year 2024 - 25 to make profit of Rs 25,00,000

Academic Year	Profit
2018-2019	1,00,000
2019-2020	2,00,000
2020-2021	3,00,000
2021-2022	3,50,000
2022-2023	4,00,000
2023-2024	5,52,000
2024-2025	0
Total	**1902000**

Go to → Data → What-If analysis → Goal Seek..



Academic Year	Profit
2018-2019	1,00,000
2019-2020	2,00,000
2020-2021	3,00,000
2021-2022	3,50,000
2022-2023	4,00,000
2023-2024	5,52,000
2024-2025	0
Total	**1902000**

Set cell → total
To value → 25 lakh
By changing cell → profit of year 2024 - 25

Academic Year	Profit
2018-2019	₹ 1,00,000.00
2019-2020	₹ 2,00,000.00
2020-2021	₹ 3,00,000.00
2021-2022	₹ 3,50,000.00
2022-2023	₹ 4,00,000.00
2023-2024	₹ 5,52,000.00
2024-2025	₹ 5,98,000.00
Total	**₹ 25,00,000.00**

It means you have to make a profit of Rs 5,98,000 to make an aggregate profit of Rs 25,00,00 by the year 2024 - 25

1. Scenario Manager : This will create different groups of values or scenarios and switch between them.

Example 1 - Room and Amount

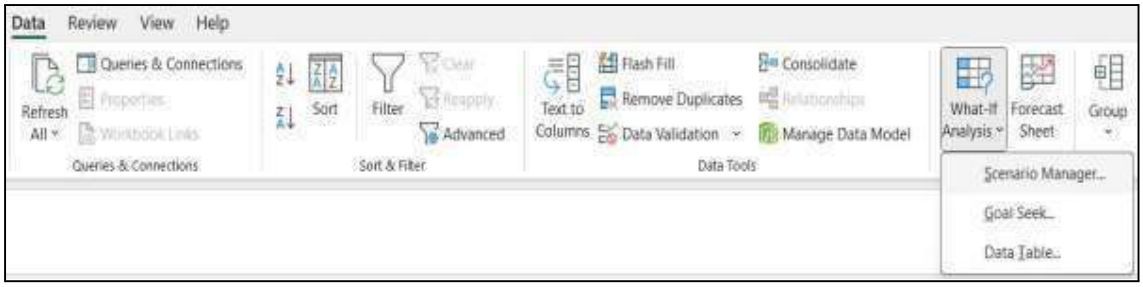
Enter the below data

Room Rate/Person	5000
No. Of Customers	50
Total Amount	

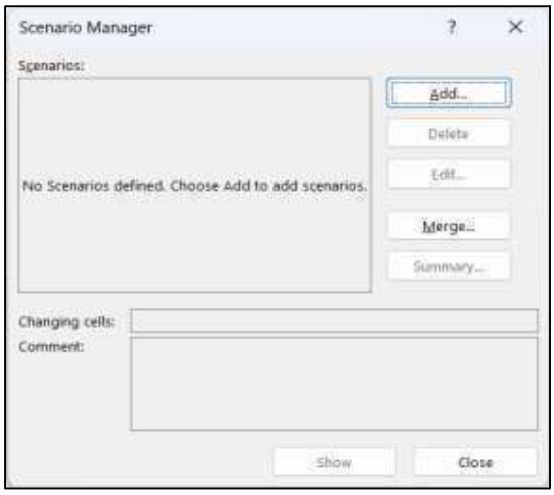
=D4*D5

C	D
Room Rate/Perso	5000
No. Of Customers	50
Total Amount	250000

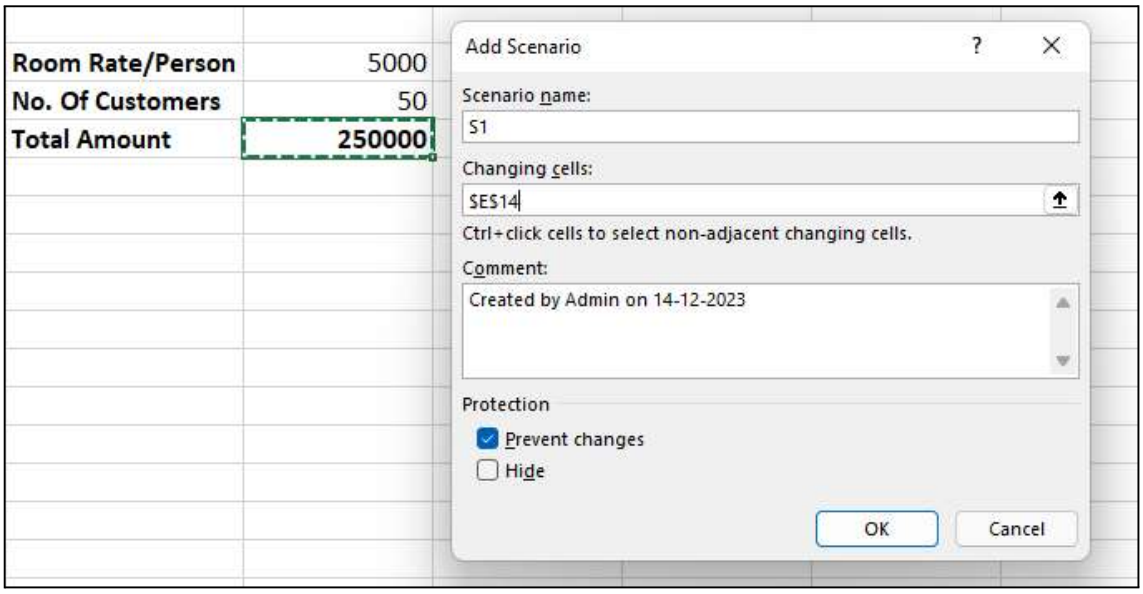
Go to
Data → What-If Analysis → Scenario

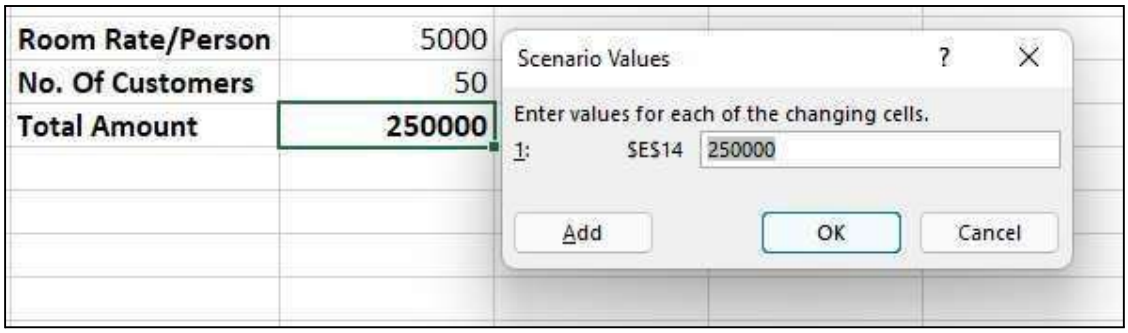


Click on Add

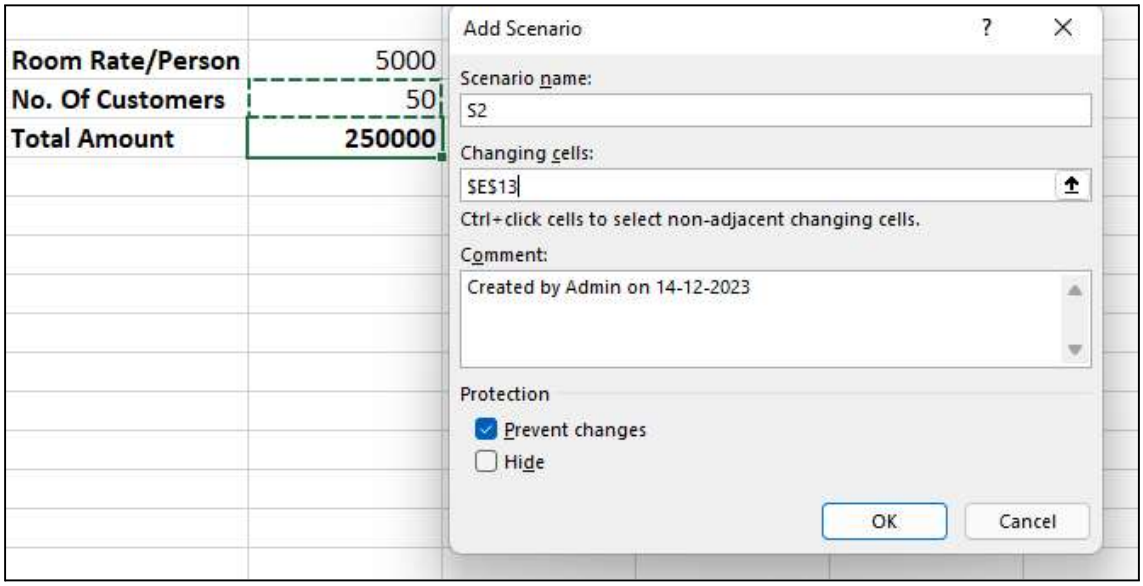


Name your scenario and changing cell in my case the no of customer will be changing cell
Click on Ok

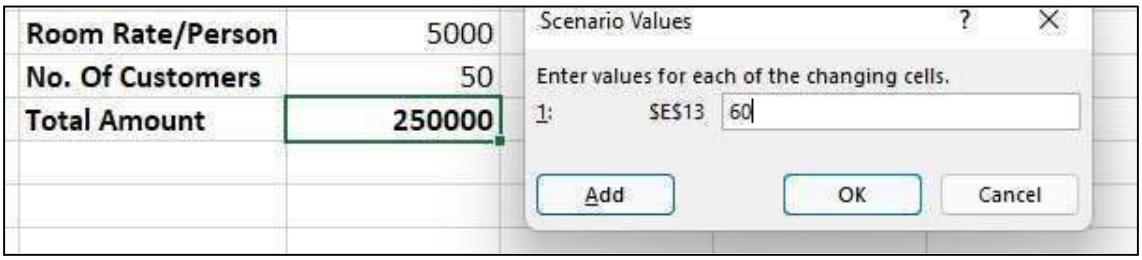




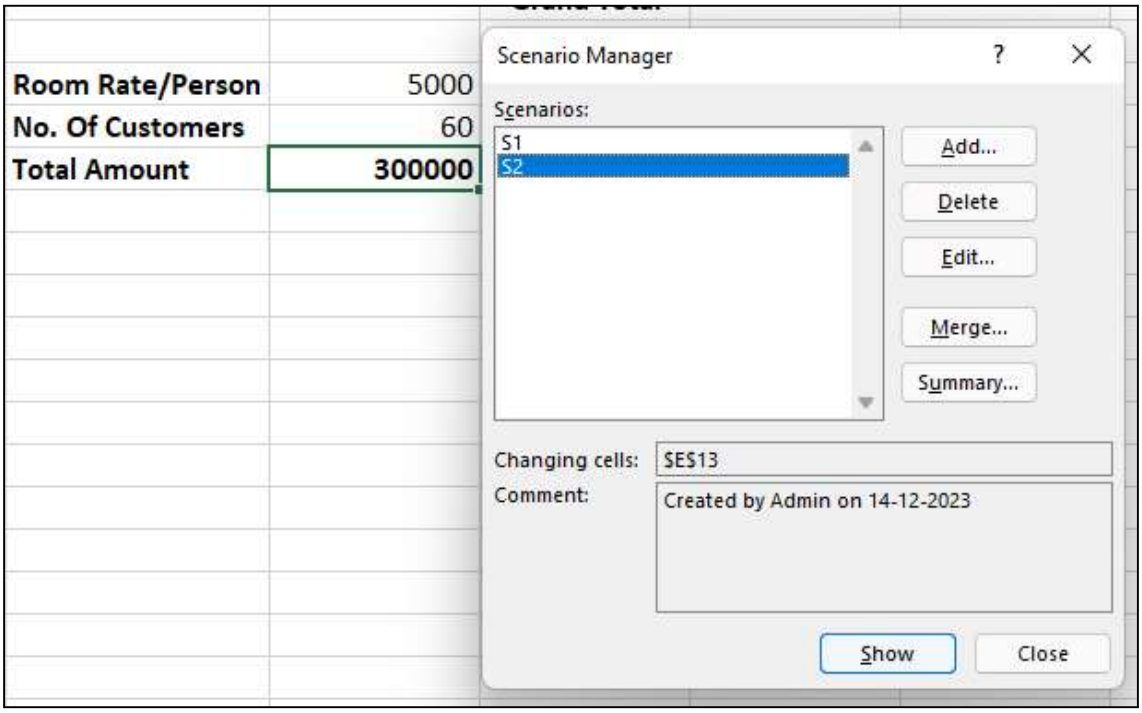
Now add second scenario S2 scenario:



Enter value for the changing cells:



Select S2 scenario then click on show button to see the changes



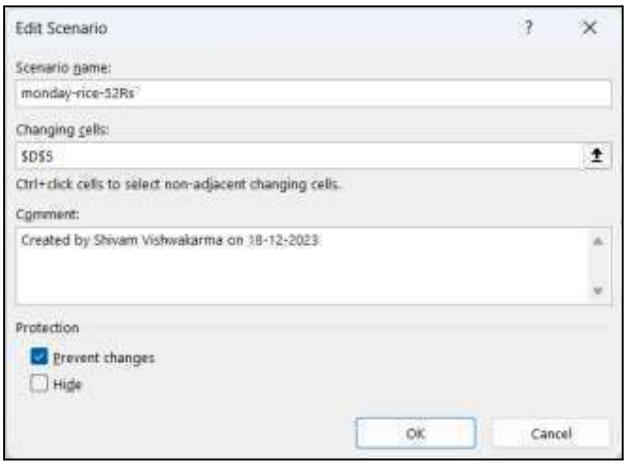
Example 2 - Grain

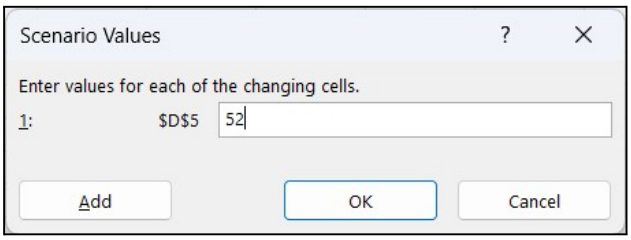
Sales Analaysis			
Grain	**Rate/Kg**	**Qty**	**Total**
Wheat	50	10	500
Rice	70	10	700
Total			1200

Total Column having formula
Rate / Qty

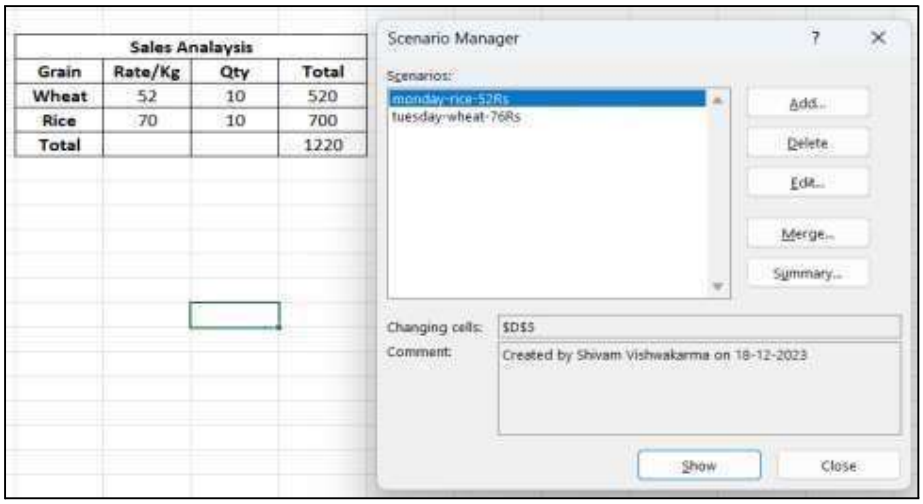
And Total Row is also having formula
Wheat's Total + Rice Total

- Add scenario manager on value of Total
- Give name to scenario manager as Monday
- where the rate of wheat will be change to 52 from 50



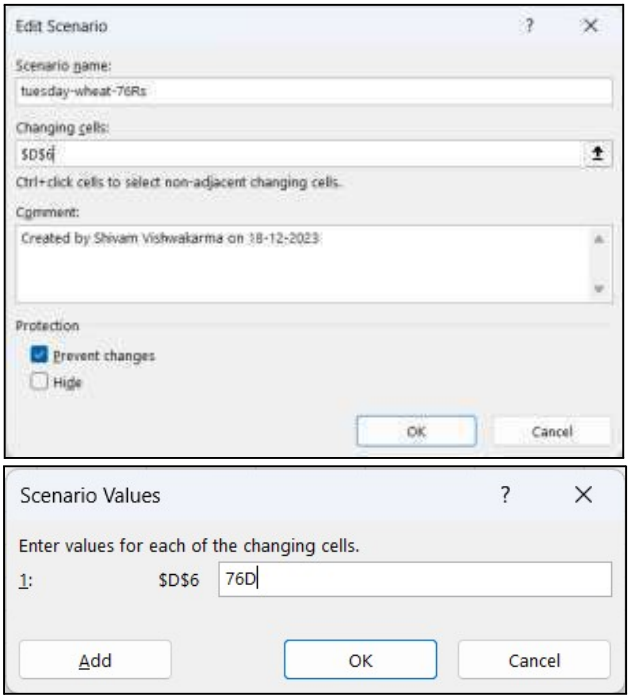


Scenario 1 Result

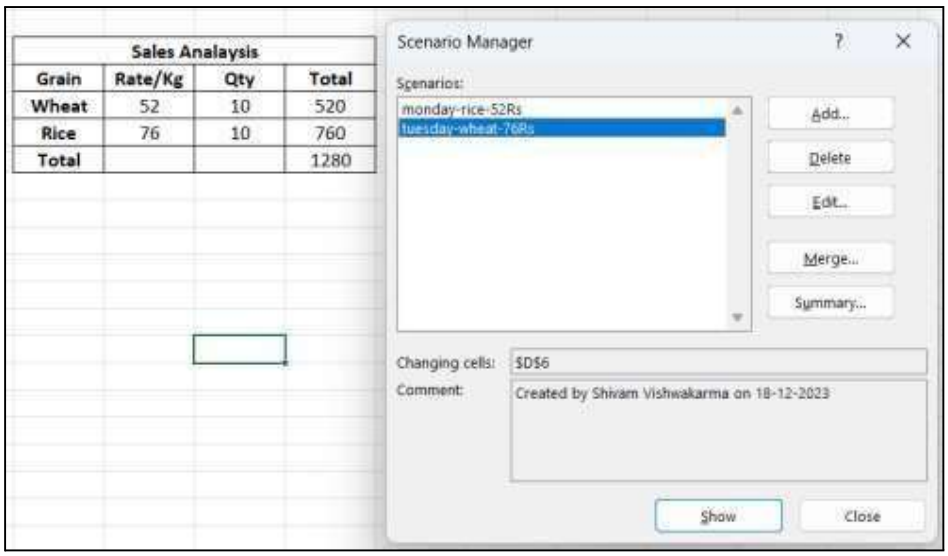


Sales Analaysis			
Grain	Rate/Kg	Qty	Total
Wheat	52	10	520
Rice	70	10	700
Total			1220

- Add second scenario named Tuesday
- where the quantity of rice will be changed to 76 from 10



Scenario 2 -Result



2. Data Table : Displays the results of multiple inputs at the same time

Example 1 - Compound Interest Calculator

Enter the below data with the formulae

- Balance = FV(B3/B4,B4*B5,,-B2)
- Interest = B6-B2

A	B
Compound Interest Calculator	
Initial Investment	2000
Annual Interest rate	5%
Compounding Periods / year	12
No. of Years	5
Balance	**₹ 2,566.72**
Interest	**₹ 566.72**

Balance

Compound Interest Calculator	
Initial Investment	2000
Annual Interest rate	5%
Compounding Periods / year	12
No. of Years	5
Balance	=FV(B3/B4, B4*B5,,-B2)
Interest	FV(rate, nper, pmt, [pv], [type

Interest

Compound Interest Calculator	
Initial Investment	2000
Annual Interest rate	5%
Compounding Periods / year	12
No. of Years	5
Balance	**₹ 2,566.72**
Interest	=B6-B2

One variable data table Example

Activity 1 → Calculate balance for different Initial Investment

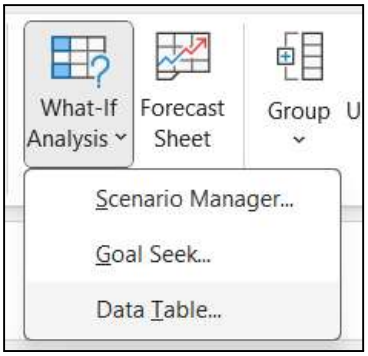
Select the cells like below manner

Initial Investment	Balance
	₹ 2,566.72
1000	
3000	
4000	
4500	

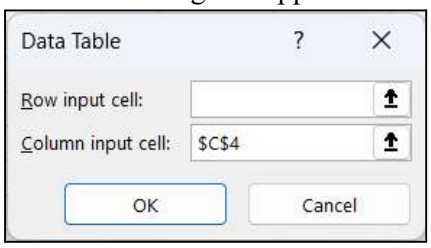
Select entire table

Initial Investment	Balance
	₹ 2,566.72
1000	
3000	
4000	
4500	

Go to
Data → What-If Analysis → Data Table



The below dialog box appear select the original initial Investment i.e 2000 click on Ok



And you will get Balance for different Initial Investment

Initial Investment	Balance
	₹ 2,566.72
1000	1000
3000	3000
4000	4000
4500	4500

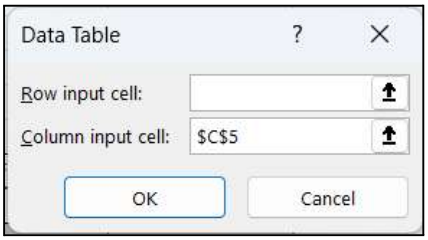
Activity 2 → Calculate balance for different Annual Rate of Interest

Select the cells like below manner

Rate	Balance
	₹ 2,566.72
6%	
7%	
8%	
9%	

Go to
Data → What-If Analysis → Data Table

Select Column input cells as the original annual rate of Investment i.e 6%



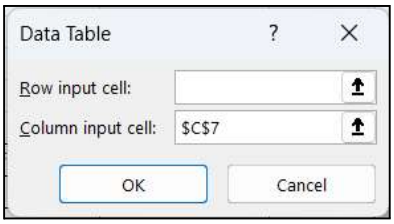
Rate	Balance
	₹ 2,566.72
6%	2697.700305
7%	2835.250519
8%	2979.691417
9%	3131.362054

Activity 3 → Calculate balance for different No. of Years
Select the cells like below manner

Year	Balance
	₹ 2,566.72
6	
7	
8	
9	

Go to
Data → What-If Analysis → Data Table

Select Column input cells as the original No. of year i.e 5



Year	Balance
	₹ 2,566.72
6	2698.04
7	2836.07
8	2981.17
9	3133.69

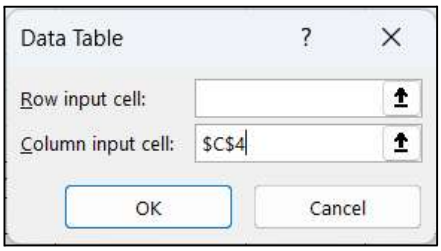
Two variable data table example

Activity 1 → Calculate Balance and Interest for Different Initial Investment
Select the cells like below manner

Initial Investment	Balanace	Interest
	₹ 2,566.72	₹ 566.72
1000		
3000		
4000		
4500		

Go to
Data → What-If Analysis → Data Table

The below dialog box appear select the original initial Investment i.e 2000 click on Ok



Two Variable Data Table		
Initial Investment	Balance	Interest
	₹ 2,566.72	₹ 566.72
1000	1283.358679	283.3586785
3000	3850.076036	850.0760355
4000	5133.434714	1133.434714
4500	5775.114053	1275.114053

Example 2 - Stationary

Enter the below data with the formula for total profit = B50*C50+B51*C51

Total No. Of Ball Pen Units	%Sold	Unit Profit	
100	60	₹ 50.00	at highest Price
100	40	₹ 20.00	at lowest Price
	Total Profit	=G37*H37+G38*H38	

Total No. Of Ball Pen Units	%Sold	Unit Profit	
100	60	₹ 50.00	at highest Price
100	40	₹ 20.00	at lowest Price
	Total Profit	₹ 3,800.00	

Practical No : 03

AIM : Perform the data classification using classification algorithm .

```
import numpy as np  classification algorithm
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score
# Load Iris dataset
iris = load_iris()

# Use only 2 features for visualization
X = iris.data[:, :2]
y = iris.target
plt.figure(figsize=(7,5))
plt.scatter(X[:, 0], X[:, 1], c=y, cmap='viridis')
plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.title("Iris Dataset Distribution")
plt.colorbar(label="Class")
plt.show()
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
cm = confusion_matrix(y_test, y_pred)
```

```
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, cmap="Blues", fmt="d",
            xticklabels=iris.target_names,
            yticklabels=iris.target_names)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
# Create mesh grid
x_min, x_max = X[:, 0].min() - 1, X[:, 0].max() + 1
y_min, y_max = X[:, 1].min() - 1, X[:, 1].max() + 1

xx, yy = np.meshgrid(
    np.arange(x_min, x_max, 0.02),
    np.arange(y_min, y_max, 0.02)
)

Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

# Plot decision boundary
plt.figure(figsize=(7,5))
plt.contourf(xx, yy, Z, alpha=0.3, cmap='viridis')
plt.scatter(X[:, 0], X[:, 1], c=y, edgecolor='k', cmap='viridis')

plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.title("Decision Boundary (Logistic Regression)")
plt.show()
```

OUTPUT :

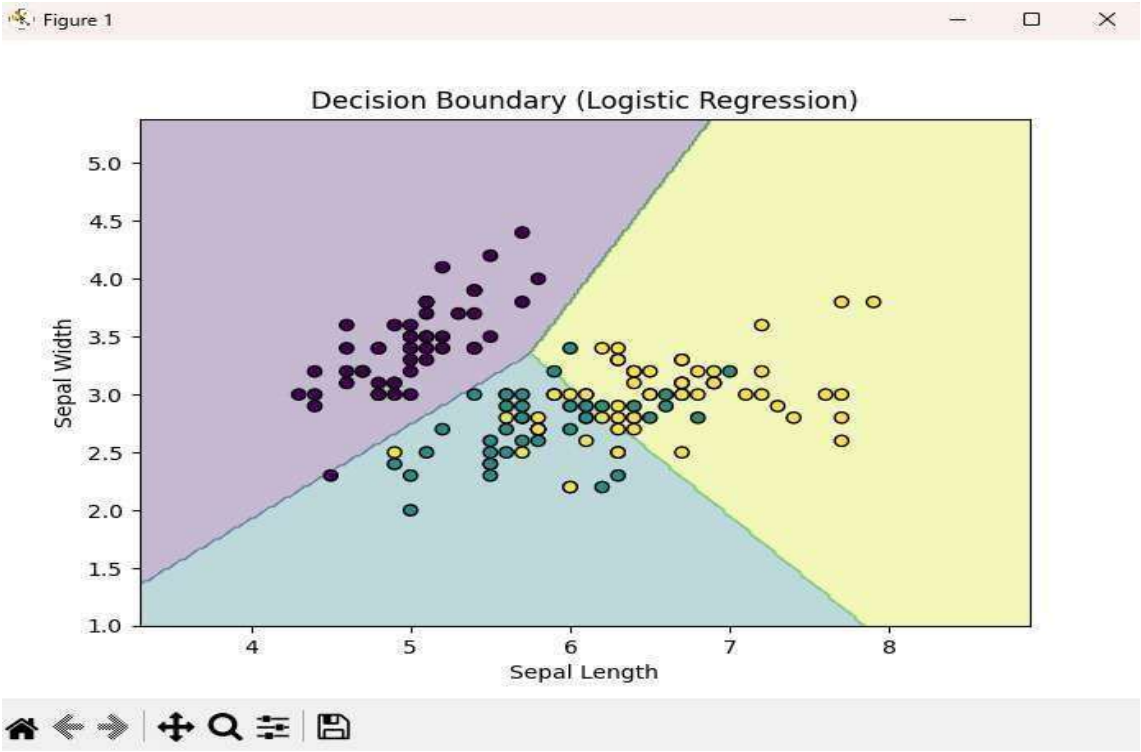


Figure 1
Decision Boundary (Logistic Regression)
Sepal Width
Sepal Length
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
4
5
6
7
8

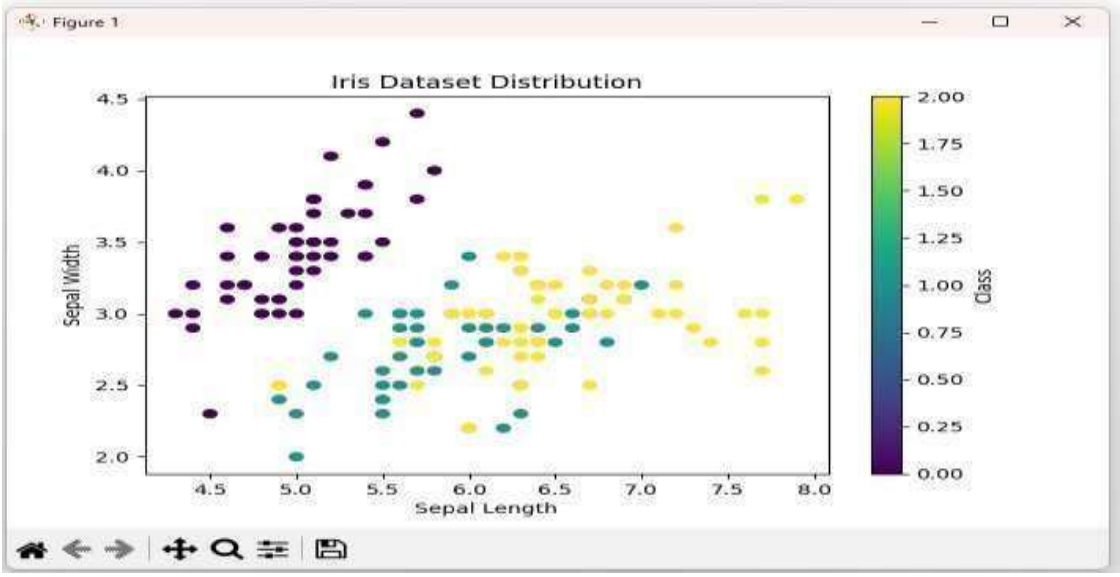
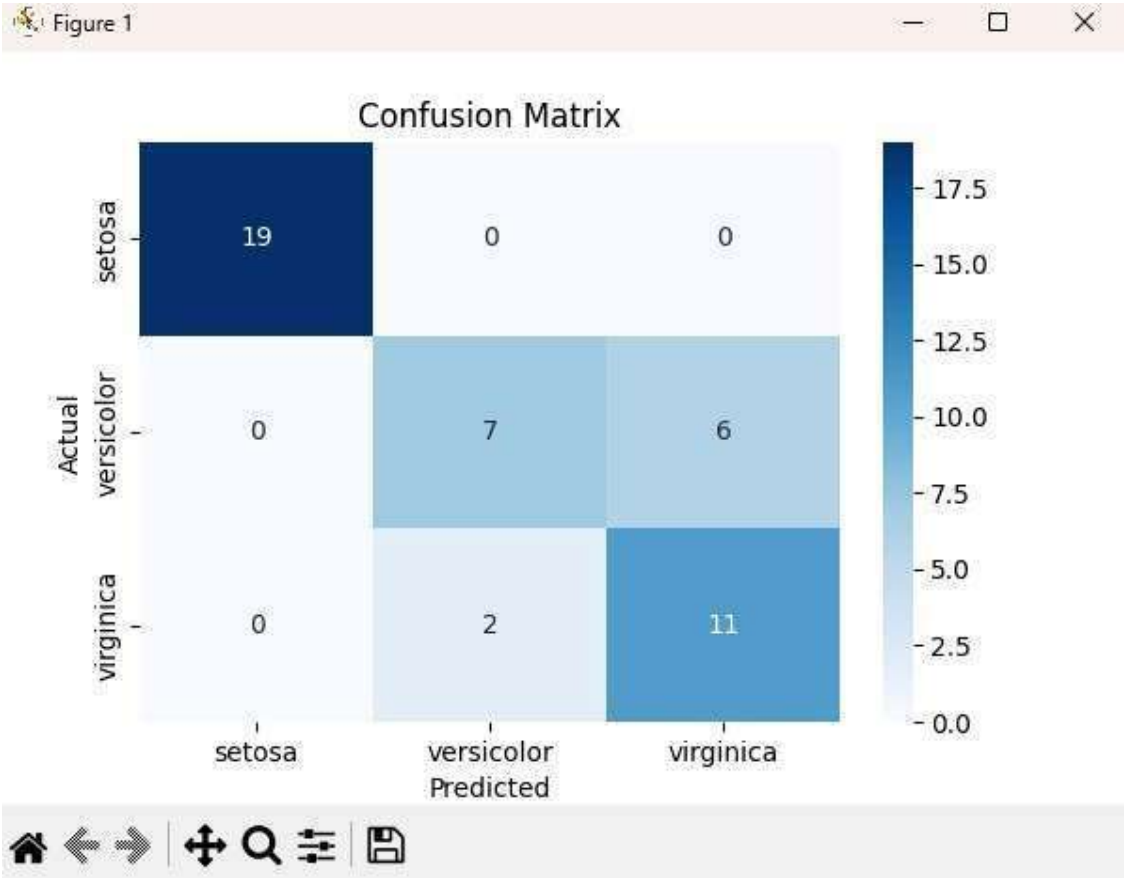


Figure 1
Iris Dataset Distribution
Sepal Width
Sepal Length
Class
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00



Accuracy: 0.8222222222222222

Practical No : 04

AIM : perform the data clustering using clustering algorithm

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.datasets import make_blobs
# Create synthetic dataset
X, y = make_blobs(
    n_samples=300,
    centers=4,
    cluster_std=1.0,
    random_state=42)
plt.figure(figsize=(7,5))
plt.scatter(X[:, 0], X[:, 1], color='gray')
plt.title("Original Data Distribution")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
kmeans = KMeans(n_clusters=4, random_state=42)
kmeans.fit(X)

labels = kmeans.labels_
centroids = kmeans.cluster_centers_
plt.figure(figsize=(7,5))
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis')
plt.scatter(centroids[:, 0], centroids[:, 1],
            s=200, c='red', marker='X', label='Centroids')

plt.title("K-Means Clustering Result")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.legend()
plt.show()
wcss = []

for k in range(1, 11):
    km = KMeans(n_clusters=k, random_state=42)
    km.fit(X)
    wcss.append(km.inertia_)

plt.figure(figsize=(7,5))
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.show()
```

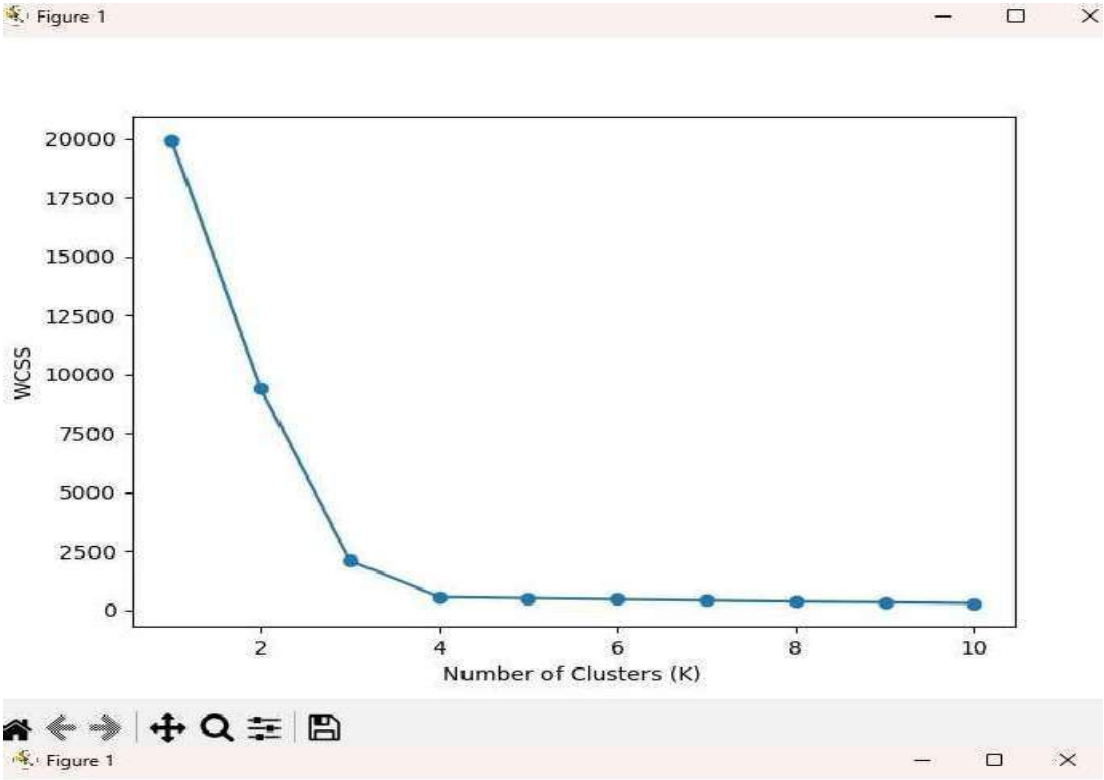


Figure 1
20000
17500
15000
12500
10000
7500
5000
2500
0
WCSS
2
4
6
8
10
Number of Clusters (K)

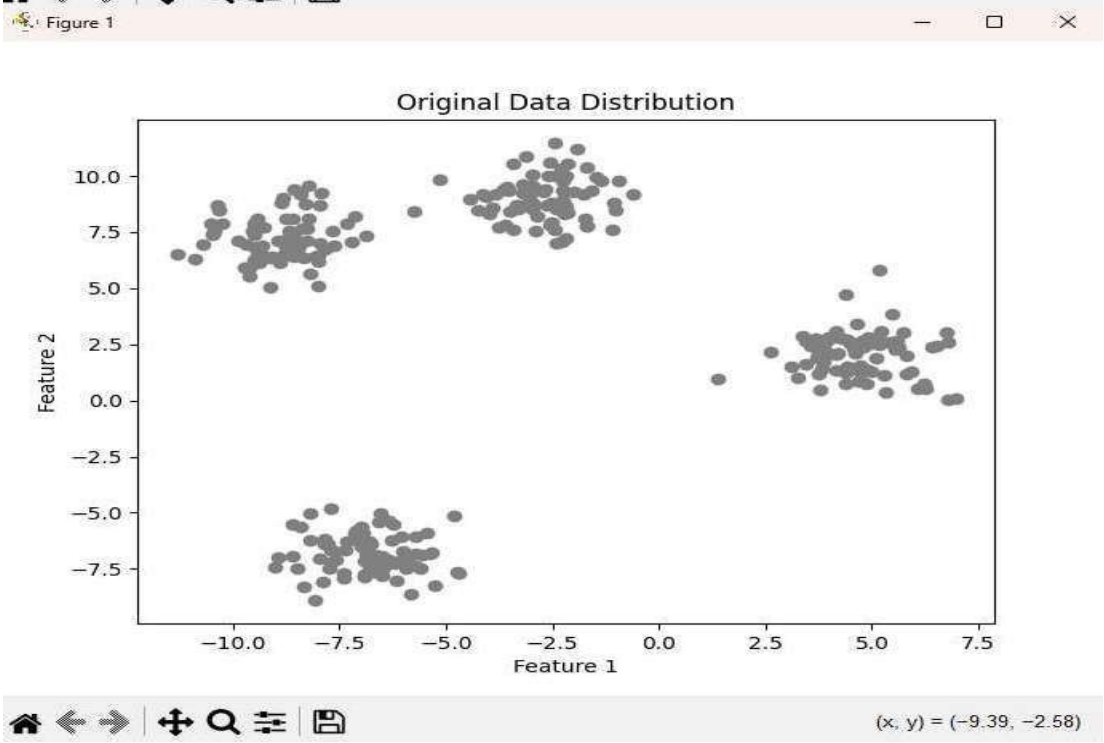


Figure 1
Original Data Distribution
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
Feature 2
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
Feature 1
(x, y) = (−9.39, −2.58)

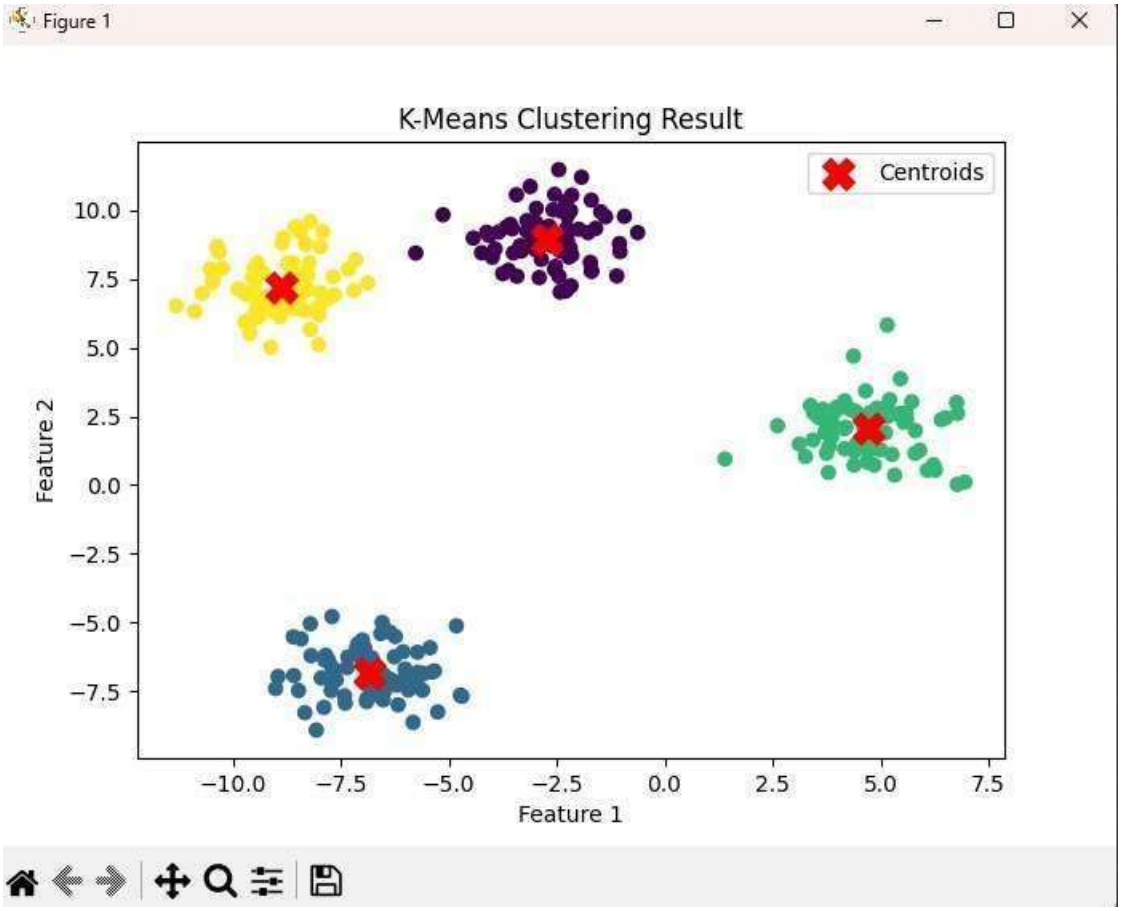


Figure 1
K-Means Clustering Result
Centroids
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
Feature 2
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
Feature 1

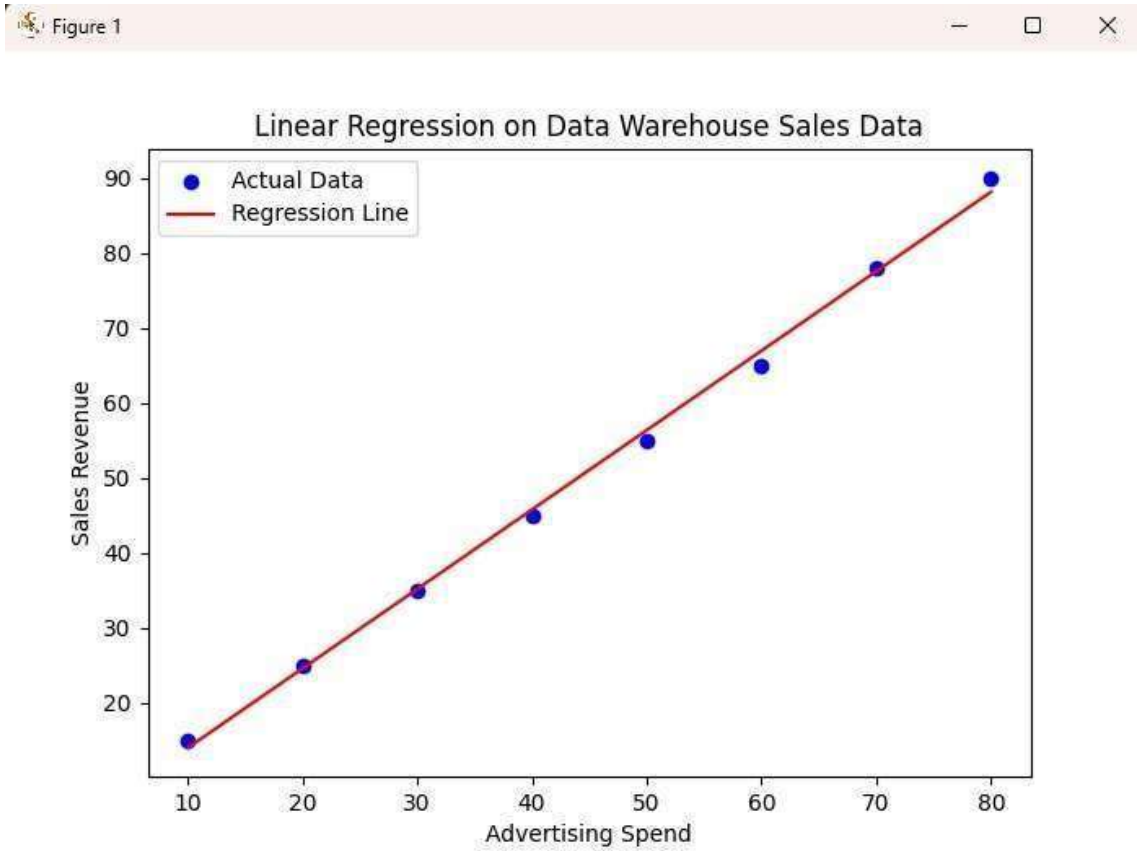
Practical No : 05

AIM : Perform the linear regression on the given datawarehouse data

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
# Create synthetic data
data = {
    "Advertising_Spend": [10, 20, 30, 40, 50, 60, 70, 80],
    "Sales_Revenue": [15, 25, 35, 45, 55, 65, 78, 90]}

df = pd.DataFrame(data)
print(df)
X = df[["Advertising_Spend"]]  # Independent variable
y = df["Sales_Revenue"]        # Dependent variable
model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)
print("Slope (m):", model.coef_[0])
print("Intercept (c):", model.intercept_)
plt.figure(figsize=(7,5))
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, y_pred, color='red', label='Regression Line')

plt.xlabel("Advertising Spend")
plt.ylabel("Sales Revenue")
plt.title("Linear Regression on Data Warehouse Sales Data")
plt.legend()
plt.show()
```



(x, y) = (19.2, 25.1)

	Advertising_Spend	Sales_Revenue
0	10	15
1	20	25
2	30	35
3	40	45
4	50	55
5	60	65
6	70	78
7	80	90

Slope (m): 1.0595238095238098

Intercept (c): 3.3214285714285623

Practical No : 06

AIM: Perform the logistic regression on the given datawarehouse data

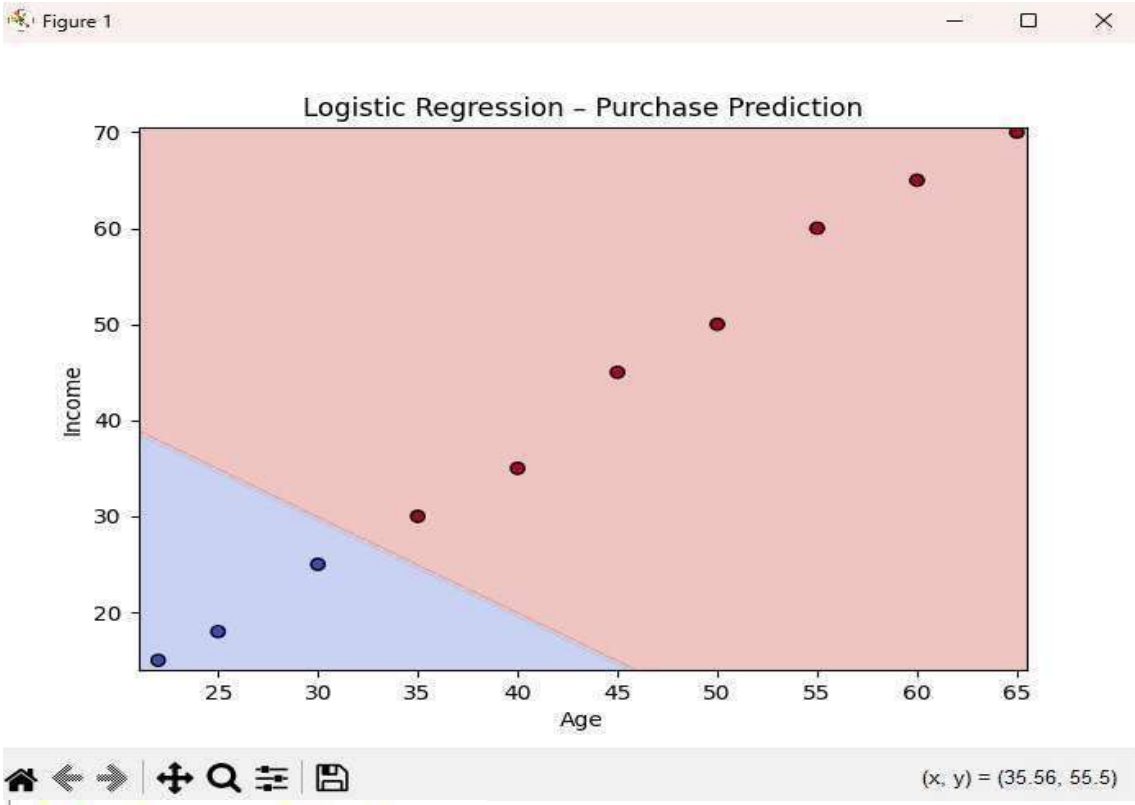
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix
# Sample synthetic data
data = {
    "Age": [22, 25, 30, 35, 40, 45, 50, 55, 60, 65],
    "Income": [15, 18, 25, 30, 35, 45, 50, 60, 65, 70],
    "Purchase": [0, 0, 0, 1, 1, 1, 1, 1, 1, 1]}

df = pd.DataFrame(data)
print(df)
X = df[["Age", "Income"]]  # Independent variables
y = df["Purchase"]          # Dependent variable
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
# Create mesh grid
x_min, x_max = X["Age"].min() - 1, X["Age"].max() + 1
y_min, y_max = X["Income"].min() - 1, X["Income"].max() + 1

xx, yy = np.meshgrid(
    np.arange(x_min, x_max, 0.5),
    np.arange(y_min, y_max, 0.5)
)

Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

# Plot
plt.figure(figsize=(7,5))
plt.contourf(xx, yy, Z, alpha=0.3, cmap="coolwarm")
plt.scatter(X["Age"], X["Income"], c=y, edgecolor='k', cmap="coolwarm")
plt.xlabel("Age")
plt.ylabel("Income")
plt.title("Logistic Regression – Purchase Prediction")
plt.show()
```



	Age	Income	Purchase
0	22	15	0
1	25	18	0
2	30	25	0
3	35	30	1
4	40	35	1
5	45	45	1
6	50	50	1
7	55	60	1
8	60	65	1
9	65	70	1

Accuracy: 1.0

Confusion Matrix:

```
[[1 0]
 [0 2]]
```

Practical No : 07

AIM: Write a Python program to read data from a CSV file, perform simple data analysis and generate basic insights

```
# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Read CSV file
file_path = "C:\\Users\\WINDOWS\\Desktop\\7.csv"
data = pd.read_csv(file_path)

# Step 2: Display first few rows
print("First 5 rows of the dataset:")
print(data.head())

# Step 3: Display basic information
print("\nDataset Information:")
print(data.info())

# Step 4: Check for missing values
print("\nMissing Values:")
print(data.isnull().sum())

# Step 5: Basic statistical analysis
print("\nStatistical Summary:")
print(data.describe())

# Step 6: Generate simple insights
print("\nBasic Insights:")
for column in data.select_dtypes(include=['int64', 'float64']).columns:
    print(f"\nColumn: {column}")
    print(f"Mean: {data[column].mean()}")
    print(f"Maximum: {data[column].max()}")
    print(f"Minimum: {data[column].min()}")

# Step 7: Data Visualization
# Plotting first numerical column
numeric_columns = data.select_dtypes(include=['int64', 'float64']).columns

if len(numeric_columns) > 0:
    data[numeric_columns[0]].plot(kind='line', title='Basic Line Plot')
    plt.xlabel("Index")
    plt.ylabel(numeric_columns[0])
    plt.show()
else:
    print("No numerical columns available for plotting.")
```

```
File Edit Shell Debug Options Window Help
= RESTART: C:/Users/WINDOWS/AppData/Local/Programs/Python/Python313/prac7.py
First 5 rows of the dataset:
   Year  ...                                Industry_code_ANZSIC06
0  2024  ...  ANZSIC06 divisions A-S (excluding classes K633...
1  2024  ...  ANZSIC06 divisions A-S (excluding classes K633...
2  2024  ...  ANZSIC06 divisions A-S (excluding classes K633...
3  2024  ...  ANZSIC06 divisions A-S (excluding classes K633...
4  2024  ...  ANZSIC06 divisions A-S (excluding classes K633...

[5 rows x 10 columns]

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55620 entries, 0 to 55619
Data columns (total 10 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Year                        55620 non-null  int64
 1   Industry_aggregation_NZSIOC 55620 non-null  object
 2   Industry_code_NZSIOC        55620 non-null  object
 3   Industry_name_NZSIOC        55620 non-null  object
 4   Units                       55620 non-null  object
 5   Variable_code               55620 non-null  object
 6   Variable_name               55620 non-null  object
 7   Variable_category           55620 non-null  object
 8   Value                       55620 non-null  object
 9   Industry_code_ANZSIC06      55620 non-null  object
dtypes: int64(1), object(9)
memory usage: 4.2+ MB
None
```

```
File Edit Shell Debug Options Window Help

Missing Values:
Year                        0
Industry_aggregation_NZSIOC 0
Industry_code_NZSIOC        0
Industry_name_NZSIOC        0
Units                       0
Variable_code               0
Variable_name               0
Variable_category           0
Value                       0
Industry_code_ANZSIC06      0
dtype: int64

Statistical Summary:
                Year
count  55620.000000
mean    2018.500000
std        3.452084
min     2013.000000
25%     2015.750000
50%     2018.500000
75%     2021.250000
max     2024.000000

Basic Insights:

Column: Year
Mean: 2018.5
Maximum: 2024
Minimum: 2013
```

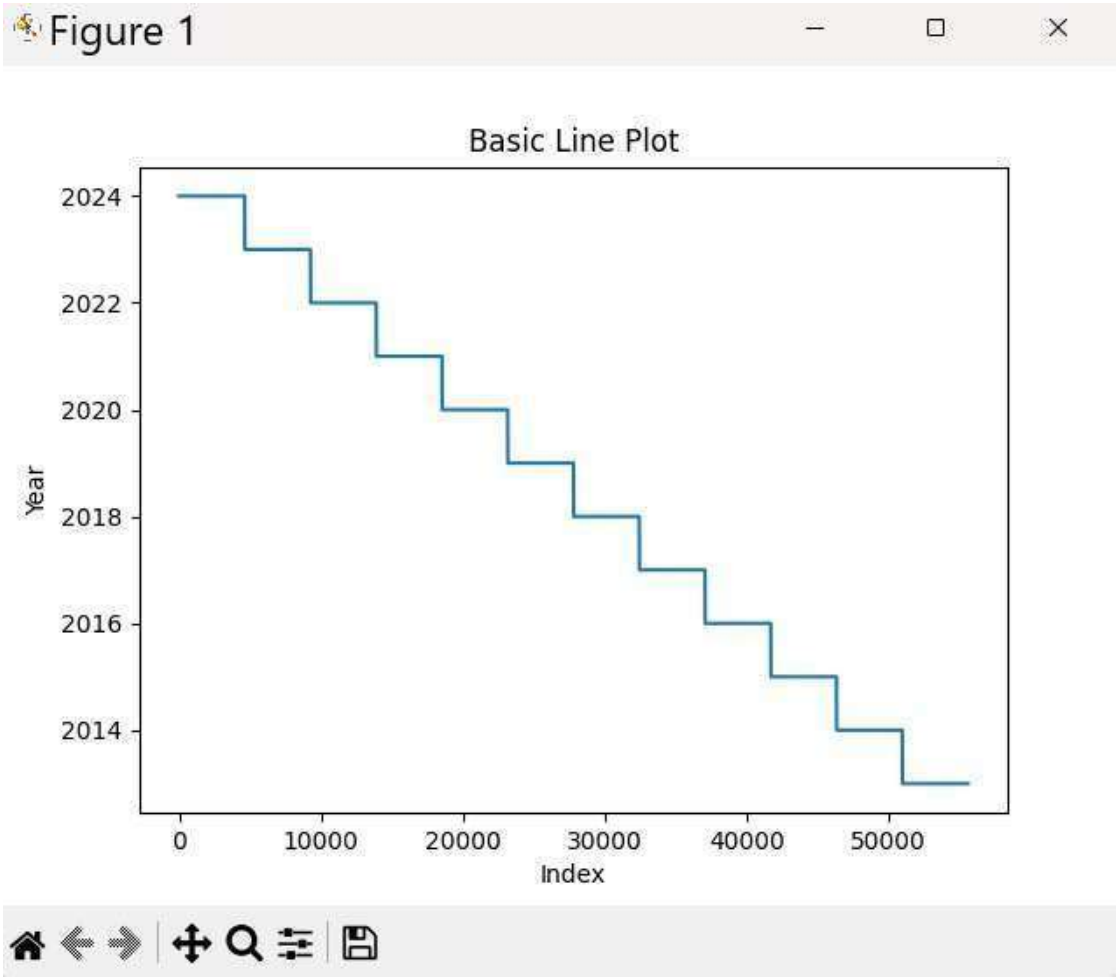


Figure 1
Basic Line Plot
2024
2022
2020
2018
2016
2014
Year
0
10000
20000
30000
40000
50000
Index

Practical No : 08

AIM: Perform data visualization using Python on any sales data

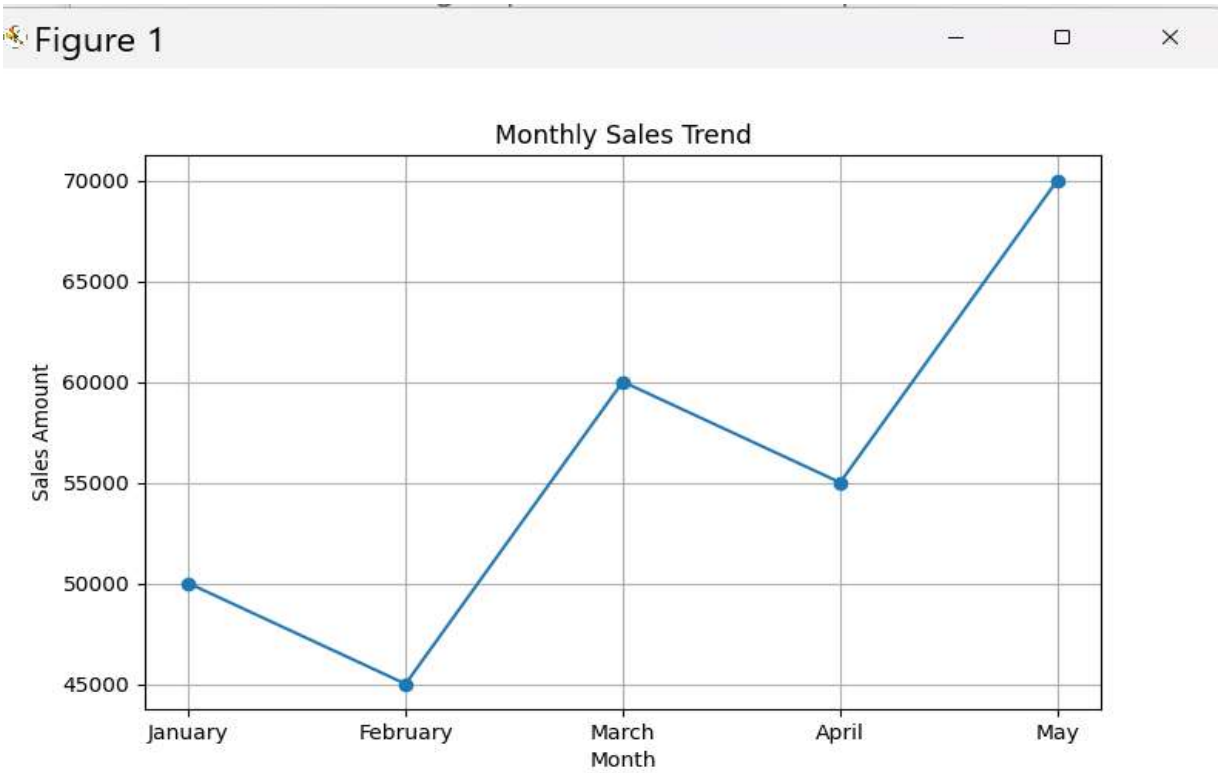
```
# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt

# Read sales data from CSV file
data = pd.read_csv("C:\\Users\\WINDOWS\\Desktop\\8a.csv")

# Display dataset
print("Sales Data:")
print(data)

# Line Chart - Monthly Sales Trend
plt.figure(figsize=(8, 5))
plt.plot(data["Month"], data["Sales"], marker='o')
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales Amount")
plt.grid(True)
plt.show()


plt.plot(data["Month"], data["Sales"], marker='o')
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```




```
== RESTART: C:/Users/WINDOWS/AppData/Local/Programs/Python/Python313/prac8a.py =
Sales Data:
      Month Product  Sales
0   January  Laptop  50000
1  February  Laptop  45000
2     March  Laptop  60000
3     April  Laptop  55000
4       May  Laptop  70000
>>>
```

Practical No : 09

AIM :

Step 1: Create Staging Database

```
CREATE DATABASE Sales_Staging;
USE Sales_Staging;
```

Step 2: Create Staging Table

```
CREATE TABLE stg_sales (
   sale_id INT,
   sale_date DATE,
   product_name VARCHAR(50),
   quantity INT,
   price DECIMAL(10,2),
   total_amount DECIMAL(10,2)
);
CREATE TABLE sales_source (
      sale_id INT PRIMARY KEY,
      sale_date DATE,
      product_name VARCHAR(50),
      quantity INT,
   price DECIMAL(10, 2)
   );
```

Step:3 insert value

```
INSERT INTO stg_sales Values (sale_id, sale_date, product_name, quantity, price,
total_amount)

 FROM sales_source;
```

Step:4 table Created

```
SELECT * FROM stg_sales
 WHERE sale_id IS NULL OR product_name IS NULL;
DELETE FROM stg_sales
   WHERE quantity <= 0 OR price <= 0;
SELECT * FROM stg_sales;
```

```
MySQL 8.0 Command Line Cli

mysql> CREATE DATABASE Sales_Stagiing;
Query OK, 1 row affected (0.01 sec)

mysql> USE Sales_Stagiing;
Database changed
mysql> CREATE TABLE stg_sales (
    -> sale_id INT,
    -> sale_date DATE,
    -> product_name VARCHAR(50),
    -> quantity INT,
    -> price DECIMAL(10,2),
    -> total_amount DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE sales_source (
    -> sale_id INT PRIMARY KEY,
    -> sale_date DATE,
    -> product_name VARCHAR(50),
    -> quantity INT,
    -> price DECIMAL(10, 2)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO stg_sales (sale_id, sale_date, product_name, quantity, price, total_amount)
    -> SELECT
    -> sale_id,
    -> sale_date,
    -> product_name,
    -> quantity,
```

```
MySQL 8.0 Command Line Cli

    -> price,
    -> quantity * price AS total_amount
    -> FROM sales_source;
Query OK, 0 rows affected (0.00 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DELETE FROM stg_sales
    -> WHERE quantity <= 0 OR price <= 0;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM stg_sales
    -> WHERE sale_id IS NULL OR product_name IS NULL;
Empty set (0.00 sec)

mysql> SELECT * FROM stg_sales;
Empty set (0.00 sec)

mysql>
```

Practical No : 10

AIM : Create the cube with suitable dimensions and facts tables based on ROLAP,MOLAP and HOLAP Models

```
CREATE DATABASE sales_dw;
CREATE TABLE dim_date (
   ->    date_id INT PRIMARY KEY,
   ->    year INT,
   ->    month INT
   -> );
USE sales_dw;
Database changed
CREATE TABLE dim_date (
   ->    date_id INT PRIMARY KEY,
   ->    year INT,
   ->    month INT
   -> );

CREATE TABLE dim_product (
   ->    product_id INT PRIMARY KEY,
   ->    product_name VARCHAR(50),
   ->    category VARCHAR(50)
   -> );

CREATE TABLE dim_customer (
   ->    customer_id INT PRIMARY KEY,
   ->    customer_name VARCHAR(50),
   ->    region VARCHAR(50)
   -> );

CREATE TABLE fact_sales (
   ->    sales_id INT PRIMARY KEY,
   ->    date_id INT,
   ->    product_id INT,
   ->    customer_id INT,
   ->    quantity INT,
   ->    sales DECIMAL(10,2),
   ->    FOREIGN KEY (date_id) REFERENCES dim_date(date_id),
   ->    FOREIGN KEY (product_id) REFERENCES dim_product(product_id),
   ->    FOREIGN KEY (customer_id) REFERENCES dim_customer(customer_id)
   -> );

INSERT INTO dim_date VALUES (1,2024,1),(2,2024,2);

INSERT INTO dim_product VALUES (101,'Laptop','Electronics'),(102,'Phone','Electronics');

 INSERT INTO dim_customer VALUES (201,'Alice','North'),(202,'Bob','South');

INSERT INTO fact_sales VALUES
   -> (1,1,101,201,2,1200.00),
```

```
    -> (2,2,102,202,1,800.00);

SELECT d.year, SUM(f.sales) AS total_sales
    -> FROM fact_sales f
    -> JOIN dim_date d ON f.date_id = d.date_id
    -> GROUP BY d.year;
```

```
mysql> SELECT d.year, SUM(f.sales) AS total_sales
    -> FROM fact_sales f
    -> JOIN dim_date d ON f.date_id = d.date_id
    -> GROUP BY d.year;
+------+-------------+
| year | total_sales |
+------+-------------+
| 2024 |     2000.00 |
+------+-------------+
1 row in set (0.01 sec)
```

```
CREATE TABLE fact_sales_summary (
    ->     year INT,
    ->     category VARCHAR(50),
    ->     total_sales DECIMAL(12,2)
    -> );

 INSERT INTO fact_sales_summary
    -> SELECT d.year, p.category, SUM(f.sales)
    -> FROM fact_sales f
    -> JOIN dim_date d ON f.date_id=d.date_id
    -> JOIN dim_product p ON f.product_id=p.product_id
    -> GROUP BY d.year, p.category;
```

```
mysql> SELECT * FROM fact_sales_summary;
+------+-------------+-------------+
| year | category    | total_sales |
+------+-------------+-------------+
| 2024 | Electronics |     2000.00 |
+------+-------------+-------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM fact_sales WHERE product_id=101;
+----------+---------+------------+-------------+----------+---------+
| sales_id | date_id | product_id | customer_id | quantity | sales   |
+----------+---------+------------+-------------+----------+---------+
|        1 |       1 |        101 |         201 |        2 | 1200.00 |
+----------+---------+------------+-------------+----------+---------+
1 row in set (0.00 sec)
```